pyAmpliCol at one loop

The recursive one-loop engine against MadLoop, Recola, OpenLoops and NJet

milestone M-QCD (2026-10-29) · engine `e18a024b` on `loop` · AMD EPYC 9754, shared host · finite part, binary64, one pinned thread, $B = 128$

×1/16 faster — ×1/4 — ×1 — ×4 — ×16 slower
run time ÷ fastest provider

Gate pass ×0.25 · fail ×1.4 · open ×1.0 · <5 ms: not gated ×1.4 — time ÷ limit
Stages inputs · numerator · scatter · NinjaRS · masters · glue
missing (why) · no colour: generation, LC · † within estimate · ¶ reference's own (DD) · ‡ user decision
ML MadLoop · Rc Recola · OL OpenLoops

Table 1: **State of the one-loop engine.** How to read it: after Figure 1.

	Process · validation	Generation	Full colour · per point, batched	Leading colour (LC)
pyAmpliCol values: the most recent measured configuration — `loop` · $2 \to 2$ to $2 \to 5$: `e18a024b`, native per-point inputs, G1 2026-10-29				
	legs · corrections · loop families; largest deviation of the finite part per reference	generate + plan export, peak memory, plan size; provider generation on this host	median per point at $B = 128$, the gate batch; provider times; gate G1 (time ÷ limit); stage shares	batched, same configuration; LC/FC; no provider ratio
$2 \to 2$	$gg \to gg$ 4 legs · QCD · 24 families (LC 12) OL·Rc·ML ≤ 6.7e−13 16 pts · median 1.8e−14 · †	**8.21 s** ×58 0.56 GiB peak · plan 0.305 MB **Rc 142 ms** · ML 9.70 s	**1.47 ms** ×0.73 $B = 128$ **ML 2.02 ms** · OL 3.20 ms · Rc 10.0 ms no gate (2→2) numerator 56 %	**0.675 ms** LC/FC 0.46 $B = 128$ · FC 1.47 ms
$2 \to 3$	$u\bar{u} \to d\bar{d}g$ 5 legs · QCD · 42 families (LC 27) ML·OL ≤ 6.7e−13 · Rc 3.3e−10 16 pts · median 8.7e−14 · ¶	**9.28 s** ×23 0.56 GiB peak · plan 0.282 MB **Rc 406 ms** · ML 9.87 s	**2.20 ms** ×1.5 $B = 128$ **OL 1.52 ms** · ML 3.17 ms · Rc 3.18 ms <5 ms: not gated ×1.1 numerator 40 % busy host: 1.88-1.91 ms in two other runs	**1.61 ms** LC/FC 0.73 $B = 128$ · FC 2.20 ms
	$u\bar{u} \to u\bar{u}g$ 5 legs · QCD · 68 families (LC 42) OL·ML ≤ 8.4e−12 · Rc 7.2e−11 16 pts · median 1.7e−13 · ¶	**8.64 s** ×19 0.56 GiB peak · plan 0.388 MB **Rc 462 ms** · ML 10.9 s	**2.60 ms** ×0.89 $B = 128$ **OL 2.92 ms** · Rc 5.55 ms · ML 8.59 ms <5 ms: not gated ×0.68 numerator 47 %	**1.51 ms** LC/FC 0.58 $B = 128$ · FC 2.60 ms
	$u\bar{u} \to e^-e^+g$ 5 legs · QCD · 11 families (LC 7) Rc·ML ≤ 1.3e−13 16 pts · median 9.6e−15 · †	**12.3 s** ×18 0.62 GiB peak · plan 0.310 MB **Rc 701 ms** · ML 14.0 s	**0.691 ms** ×0.73 $B = 128$ **ML 0.948 ms** · Rc 1.08 ms · OL no library <5 ms: not gated ×0.73 numerator 39 %	**0.652 ms** LC/FC 0.94 $B = 128$ · FC 0.691 ms
	$u\bar{u} \to ggg$ 5 legs · QCD · 90 families (LC 62) ML·Rc ≤ 1.5e−12 · OL 8.4e−11 16 pts · median 6.7e−14 · ¶	**9.28 s** ×15 0.57 GiB peak · plan 0.656 MB **Rc 634 ms** · ML 17.6 s	**5.26 ms** ×0.42 $B = 128$ **OL 12.4 ms** · Rc 25.3 ms · ML 28.2 ms pass ×0.33 numerator 58 %	**2.69 ms** LC/FC 0.51 $B = 128$ · FC 5.26 ms
	$gg \to ggg$ 5 legs · QCD · 154 families (LC 77) ML·Rc·OL ≤ 1.0e−6 16 pts · median 4.9e−13 · †	**10.2 s** ×1.5 0.62 GiB peak · plan 1.85 MB **Rc 6.63 s** · ML 47.0 s	**19.5 ms** ×0.20 $B = 128$ **OL 97.0 ms** · ML 247 ms · Rc 351 ms pass ×0.15 numerator 59 %	**8.05 ms** LC/FC 0.41 $B = 128$ · FC 19.5 ms
$2 \to 4$	$u\bar{u} \to e^-e^+gg$ 6 legs · QCD · 69 families (LC 33) Rc·ML ≤ 9.4e−11 16 pts · median 6.3e−14 · †	**13.1 s** ×16 0.63 GiB peak · plan 0.624 MB **Rc 833 ms** · ML 24.8 s	**4.93 ms** ×0.48 $B = 128$ **Rc 10.2 ms** · ML 27.3 ms · OL no library <5 ms: not gated ×0.37 numerator 58 %	**1.73 ms** LC/FC 0.35 $B = 128$ · FC 4.93 ms
	$gg \to t\bar{t}gg$ 6 legs · QCD · 1 086 families (LC 375) OL·Rc·ML ≤ 1.4e−8 16 pts · median 1.9e−11 · †	**14.4 s** ×0.23 0.72 GiB peak · plan 3.71 MB **Rc 61.4 s** · ML 218 s	**92.2 ms** ×0.20 $B = 128$ **OL 472 ms** · Rc 1.97 s · ML 5.45 s pass ×0.15 numerator 49 %	**31.6 ms** LC/FC 0.34 $B = 128$ · FC 92.2 ms
	$u\bar{u} \to gggg$ 6 legs · QCD · 628 families (LC 375) Rc 9.3e−12 · OL 1.6e−10 16 pts · median 1.5e−12 · †	**14.4 s** ×0.44 0.72 GiB peak · plan 4.54 MB **Rc 32.8 s**	**73.4 ms** ×0.20 $B = 128$ **OL 374 ms** · Rc 908 ms · ML not exported pass ×0.15 numerator 65 %	**29.0 ms** LC/FC 0.40 $B = 128$ · FC 73.4 ms
	$gg \to gggg$ 6 legs · QCD · 1 100 families (LC 550) Rc·OL·ML ≤ 2.2e−7 8 pts · median 3.0e−11 · †	**31.9 s** ×0.015 1.27 GiB peak · plan 16.1 MB **Rc 35.4 min** · ML 144.6 min	**384 ms** ×0.14 $B = 128$ **OL 2.84 s** · Rc 17.6 s · ML 38.7 s pass ×0.10 numerator 56 %	**176 ms** LC/FC 0.46 $B = 128$ · FC 384 ms

	Process · validation	Generation	Full colour · per point, batched	Leading colour (LC)
pyAmpliCol values: the most recent measured configuration — `loop` · $2 \to 2$ to $2 \to 5$: e18a024b, native per-point inputs, G1 2026-10-29				
	legs · corrections · loop families; largest deviation of the finite part per reference	generate + plan export, peak memory, plan size; provider generation on this host	median per point at $B = 128$, the gate batch; provider times; gate G1 (time ÷ limit); stage shares	batched, same configuration; LC/FC; no provider ratio
$2 \to 5$	$u\bar{u} \to e^-e^+d\bar{d}g$ 7 legs · QCD · 242 families (LC 114) ML·OL·Rc ≤ 5.7e−11 16 pts · median 8.0e−13 · †	**17.1 s** ×2.5 0.66 GiB peak · plan 1.13 MB **Rc 6.73 s** · ML 58.6 s	**16.8 ms** ×0.48 **Rc 35.3 ms** · OL 51.0 ms · ML 132 ms $B = 128$ pass ×0.37 numerator 54 %	**6.41 ms** LC/FC 0.38 $B = 128$ · FC 16.8 ms
	$u\bar{u} \to e^-e^+ggg$ 7 legs · QCD · 463 families (LC 164) ML·Rc·OL ≤ 1.1e−10 16 pts · median 1.5e−12 · †	**17.1 s** ×1.3 0.68 GiB peak · plan 2.45 MB **Rc 13.4 s** · ML 146 s	**58.1 ms** ×0.23 **Rc 251 ms** · OL 346 ms · ML 1.42 s $B = 128$ pass ×0.18 numerator 69 %	**12.4 ms** LC/FC 0.21 $B = 128$ · FC 58.1 ms
	$gg \to t\bar{t}e^-e^+g$ 7 legs · QCD · 579 families (LC 164) Rc·OL·ML ≤ 8.7e−9 16 pts · median 2.7e−11 · †	**17.9 s** ×0.65 0.75 GiB peak · plan 2.13 MB **Rc 27.7 s** · ML 165 s	**63.1 ms** ×0.16 **OL 404 ms** · Rc 620 ms · ML 3.04 s $B = 128$ pass ×0.12 numerator 63 %	**11.6 ms** LC/FC 0.18 $B = 128$ · FC 63.1 ms
	$gg \to t\bar{t}e^-e^+g$ 7 legs · EW · families not recorded **in development (WP3.2, M-SM): EW native inputs (EWI-N), the EW ladder and the target (EWI-7, EWI-9)**	***in development (WP3.2, M-SM)*** EW native inputs (EWI-N), the EW ladder and the target (EWI-7, EWI-9): no configuration generates it yet **Rc 21.9 s** · ML 10.9 min	***in development (WP3.2, M-SM)*** production, s/pt (20 Oct): OL 0.96–1.12 · Rc 0.95–1.02 · ML 6.90–7.08 gate limit 1.235 s/pt (Recola)	***in development (WP3.2, M-SM)***
	$u\bar{u} \to gggg$ 7 legs · QCD · 5 258 families (LC 2 712) ML 1.9e−12 · Rc 2.9e−10 3 pts · median 7.1e−11 · ¶	**138 s** ×0.014 3.02 GiB peak · plan 69.2 MB **Rc 170.3 min** · ML 12.2 h	**1.75 s** ≥×0.0096 **Rc ≤182 s** · ML 569 s · OL no library $B = 128$ pass ×0.0031 numerator 65 %	**555 ms** LC/FC 0.32 $B = 128$ · FC 1.75 s
	$gg \to gggg$ 7 legs · QCD · 4 707 families (LC 4 707) NJet DD 4.4e−8 10 pts · median 3.2e−9 · † · top-free (particle_veto = [t]), Nf = 5 ‡ pts 4, 8 above 2.0e−8: accepted (user 2026-10-25)	**383 s** 9.54 GiB peak · plan 160 MB ML diagrams 17.4 h, stopped in export · Rc unfinished	**7.04 s** $B = 128$ no gate (no matched provider) numerator 51 %	**3.80 s** LC/FC 0.54 $B = 128$ · FC 7.04 s

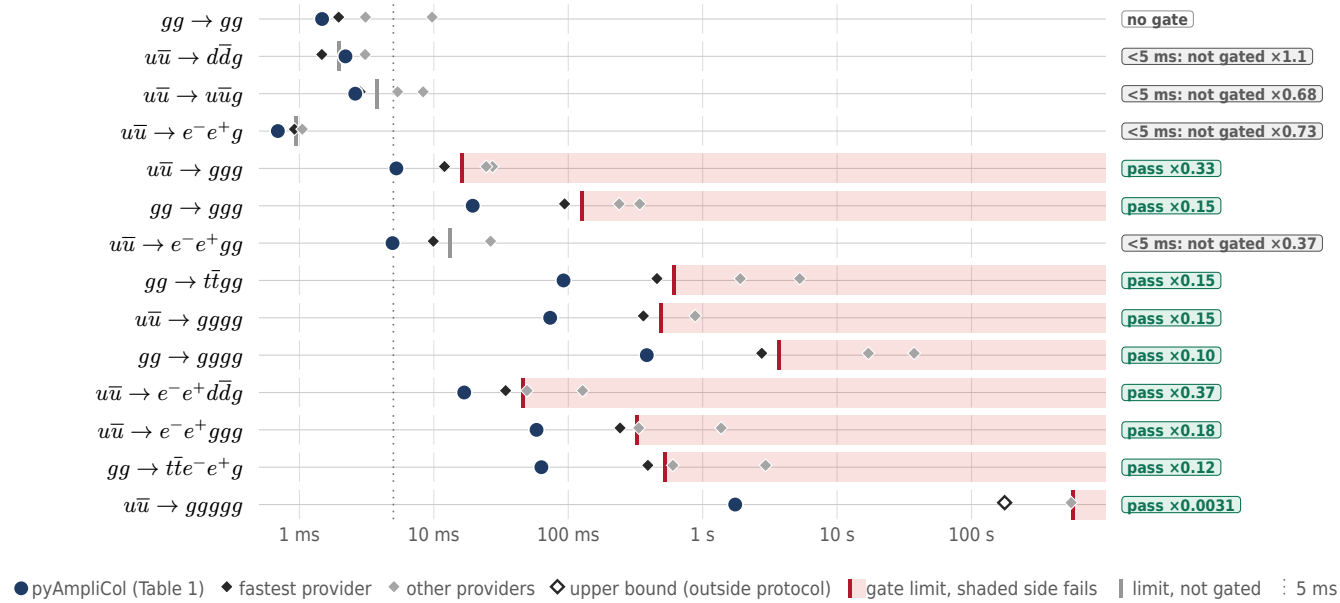


Figure 1: **Batched full colour against the gate** (finite part, $B = 128$; the rows of Table 1 with a matched provider).

Reading the summary

Table 1 has one row per test process, by multiplicity. A cell gives pyAmpliCol's value in bold and, at its right, the multiplier *pyAmpliCol ÷ the fastest provider* on this host (in bold among the provider values). A provider time marked ≤ ($u\bar{u} \to gggg$) comes from a run outside the protocol that does the production work and more (Recola's reference run evaluates the IR poles too), so it bounds that provider's production time from above, and the multiplier against it is a lower bound (≥); it is never a gate limit. Only run times are coloured: green faster, white at parity, red slower ($\log_2$ scale, clamped at ×1/16 and ×16). Generation cells carry their multiplier uncoloured (generation time is not the main metric, user 2026-10-24); leading colour has no provider ratio. Grey cells say why a value is missing. Run times are per phase-space point in batches, finite part, stability checks off; providers in production mode (Section 1). Every value is the **most recent measured configuration** of its cell (by git ancestry of the revisions; a newer check that only repeats part of an older comparison does not replace it): $2 \to 2$ to $2 \to 5$: the engine on `loop` with native per-point inputs, gate G1 at `e18a024b` (2026-10-29). The engine before G-int (Python per-point inputs, G1 at `1c971289`) is history in Section 3. The column header of Table 1 names these configurations once.

Gate chips (Table 1 and Figure 1): the time over the binding limit (MadLoop's time, or 1.30 × the faster of Recola and OpenLoops), **pass** or **fail** when the whole uncertainty band is below or above 1, else **open**. The user's gate rule of 2026-10-24: a process whose time per point at $B = 128$ is below 5 ms has no blocking gate, its verdict is "below 5 ms: not gated (user 2026-10-24)" and its chip ("<5 ms: not gated") keeps the measured ratio ($u\bar{u} \to d\bar{d}g$ 2.20 ms, 1.12 of its limit, $u\bar{u} \to u\bar{u}g$ 2.60 ms, 0.68 of its limit, $u\bar{u} \to e^-e^+g$ 0.691 ms, 0.73 of its limit and $u\bar{u} \to e^-e^+gg$ 4.93 ms, 0.37 of its limit). $gg \to gg$ ($2 \to 2$) has no gate. The gate is taken at $B = 128$ only (user 2026-10-28, `gate_decisions.gate_batch`): $B = 1$ and the $2 \to 5$ rows' $B = 16$ are measured and **reported** with their ratios (Table 4, Table 5), without a verdict or a verdict colour.

Stage shares (the bar of the full-colour cell; the largest is named; Table 6): inputs = trees, Born and counterterm inputs (on `loop` the executor's own timer of its native trees, since lc-perf); glue = point parameters, input conversion, assembly and the rest of the call. Times use the protocol's schedule B.

References and deviations. ML MadLoop 3.7.0, Rc Recola 2.2.5 (COLLIER), OL OpenLoops 2.1.5 (prebuilt libraries: no generation time), NJet DD NJet 3.1.2 in double-double. Deviation $|a - b| \,/\, \max(|a|, |b|, |B|\ \alpha_s/2\pi)$ of the binary64 finite part at the timed points (median: over the points, the largest over the references); † the largest lies within the binary64 stability estimate of the same configuration at that point; ¶ the largest is the reference's own: pyAmpliCol's double-double value at that point lies as far from it, and its binary64 value lies within E of the double-double one. ‡ $gg \to ggggg$: the reference check's points 4 and 8 (4.4×10^{-8} and 2.1×10^{-8} from NJet DD) exceed its tolerance 2×10^{-8} in binary64: **accepted (user 2026-10-25): O(1e-8) binary64, within the estimate, DD = NJet** (estimates E 1.7×10^{-6} and 3.3×10^{-7}; pyAmpliCol's double-double values equal the reference's); the check keeps its tolerance and, asserting double-double at these points since the user's E1 ruling (2026-10-29), passes (`data/milestone.json validation_decisions`).

Configurations (main-configuration rule of `scripts/build_summary.py`; `data/summary.json` names the configuration and source of every value). $2 \to 2$ to $2 \to 5$: the engine on `loop`, gate G1 at `e18a024b`, build `loop-native-v2-slice-d0fef168`, native per-point inputs (Section 3). Generation excludes model loading; unbatched development timings (e.g. a seven-gluon point with its rotation check) are not in this table (Section 5). G1 generates every row cold since milestone m1026a (`11542fd5`), in its own process with a fresh cache directory, so the counterterm caches are rebuilt in every run (Section 3, generation protocol).

Notes under a value (`data/milestone.json cell_notes`) qualify it and leave it as measured, in amber a known limitation of the value, in grey its run conditions: $u\bar{u} \to d\bar{d}g$, full colour ("busy host: 1.88-1.91 ms in two other runs"): 2.20 ms per point (1.12 of OpenLoops's limit, below 5 ms: not gated) against 1.83 ms at m1028a; the same engine and native ran it at 1.88 ms (smoke run) and 1.91 ms (public path); not repeated (one timing round, user 2026-10-28).

Figure 1 places each row's batched full-colour value of Table 1 (dots; the most recent measured configuration: $2 \to 2$ to $2 \to 5$: `e18a024b`, native per-point inputs, G1 2026-10-29) against the providers (diamonds, the darkest the fastest) and the gate limit (red bar, the shaded side fails; grey bar: the limit of a process below 5 ms, not gated; dotted line: 5 ms), for the rows with a matched provider, at $B = 128$. The ratios and bands at every batch size: Table 5.

1 Status at this milestone: M-QCD

On `loop` (engine e18a024b; merges since de18f169, engine 28a16255)

`6e123fdb`	**onshell-threshold**	Threshold lanes to the generic engine	`01e80d7c`	**MQ-VALIDATE**	M-QCD validation campaign
`1acb393f`	**fail-closed-2**	Vanishing couplings leave loops	`d8d9349f`	**MQ-NC**	Shipped N_c declaration
`5301e24a`	**glibc-floor**	Typed glibc-floor error	`6b78dc3b`	**MQ-5F**	Five flavours, massive b
`d3081568`	**MQ-WARD**	Generic gauge-Ward check	`d525a09c`	**MQ-EWJOIN**	q q~ > V V g, Catani sectors
`da6980b0`	**MQ-FFT**	Certified FFT seed contraction	`e4964fc3`	**MQ-SELECT**	LC helicity, colour-flow selection
`141fe125`	**MQ-REF**	M-QCD reference campaign	`539dffdd`	**MQ-LEGACY-REMOVE**	v2 the only one-loop engine
`321efdb4`	**MQ-MODEL**	Stock loop_sm on the public route	`e18a024b`	**g1-accuracy**	G1 accuracy per the E1 ruling
`a2aa0d84`	**MQ-STAB**	Recommended stability checks			

Before (engine 28a16255): v2 waves 0-1 (gates S1, G0), native executor, colour, stability API, schedule M, observables, NJet reference, public v2 engine with plan v4 artifacts, counterterms as plain tree insertions (addendum J), runtime pole selection, Rust A0-A3, the G1 harness, G-int (native per-point inputs and generation stages), the summary-first report; at milestone m1025a ct-perf, ew-poles (EW guards, QED Catani operator), lc-perf, artifact-portability and ew-lines; at m1025c native-dag A4, ew-gauge EWI-4a, precision-ap, ew-ct EWI-0/EWI-3, measure-g1-7leg (G1 at 2->5), reference-relocate, model-determinism, native-lowering D phase 1 part 1 and onshell-consistency; at m1026a ew-gauge EWI-4b, native-seeds B phase 1, native-cells C phase 1, genericity-fixes, ew-poles EWI-6b, fermion-lines and generation-scale-2; at m1028a ew-ct EWI-3R, ew-epsilon EWI-7b, fail-closed, sdk-parity, ew-leptons EWI-5, identical-flavour-lc and gate-b128. **Native build** `loop-native-v2-slice-d0fef168` (digest `1be09f92`…).

M-QCD (full QCD one-loop support on the stock `loop_sm`, reproducing MadLoop; user plan 2026-10-28; `m-qcd/PLAN.md` §5)

- **Validation campaign** at `e18a024b` (Section 2, Table 21): 78 rows in 9 classes, 76 pass, 1 partial, 1 fail (F7, F11, accepted limitations: Missing box) no discrepancy above 10^{-5}, no wrong pole or sector. Largest deviations: DD from the reference 1.2×10^{-11} (A18), binary64 from DD 2.3×10^{-10} (C05), poles from Catani in DD 1.2×10^{-22} (A09).
- **Public route** on the shipped `loop_sm` (`Generator`, `Runtime.load`, `evaluate`): one row per class passes, DD from MadLoop ≤ 3.6×10^{-13}. Wheel self-test in two fresh venvs (PyPI, Community Symbolica): all checks pass its virtual: u u~ > e- e+ / Z [virt=QCD] on the shipped loop_sm-no_b_mass (full colour): Born and Laurent coefficients = MadLoop to 2.8e-13 at 3 points.
- **Colour:** full colour is the loop default; LC and NLC (shipped N_c declaration) pass rule 7 on 13 rows and the smoke on every shipped restriction; LC helicity and colour-flow selection. **FFT** seed contraction = direct to rounding; its cost: ×0.46–5.9 of direct, the whole evaluation ×0.93–1.29 (Table 3).
- **Ward** (one gluon's $\varepsilon \to p$, native inputs): passes up to seven gluons (passed (MQ-WARD)). **R3 for QCD** recorded (DECISIONS 2026-10-29 ~18:25; deferred: provider protocol-mode columns, M-FINAL).

Gate G1 (Section 3; finite part, $2 \to 3$ and beyond)

- **Most recent configuration** (Table 1 and Figure 1): native per-point inputs on `loop` (G1 at `e18a024b`) at $B = 128$: 9 of 9 gated processes pass; $u\bar{u} \to d\bar{d}g$, $u\bar{u} \to u\bar{u}g$, $u\bar{u} \to e^-e^+g$ and $u\bar{u} \to e^-e^+gg$ run below 5 ms per point: **not gated** (user 2026-10-24; 1.12, 0.68, 0.73, 0.37 of the limit). $B = 1$ (16 lanes, the runs of `28a16255`, not re-timed) is **reported**, not gated (user 2026-10-28): ×1.0–5.3 of the limit (Table 5).
- **$2 \to 5$** (full and leading colour at $B = 128$; $B = 16$ reported): 4 of 4 gated rows pass ($u\bar{u} \to e^-e^+d\bar{d}g$ 16.8 ms, 0.37 of its limit; $u\bar{u} \to e^-e^+ggg$ 58.1 ms, 0.18 of its limit; $gg \to t\bar{t}e^-e^+g$ 63.1 ms, 0.12 of its limit; $u\bar{u} \to ggggg$ 1.75 s, 0.0031 of its limit, ≥×0.0096 against Recola's upper bound); $gg \to ggggg$ 7.04 s at full and 3.80 s at leading colour per point, no matched provider and no gate.
- **Before G-int** (Python per-point inputs, G1 at `1c971289`; history, Section 3): failed every gated process it generated, at ×11–20 of the binding limit at $B = 128$.

Missing or incomplete (typed refusal keys, listed and shown by the dry run)

- **Refused, out of M-QCD:** $1 \to n$ decays (`decay-process`), MG5 decay chains and `$` (`madloop-syntax`); loop-induced requests (`loop-induced`, after M-SM); a coloured particle with a width in a tree current (`coloured-width`; recipes `loop_sm-no_widths` or `fixed_parameters = {WT = "0"}`); Majorana fermions (`majorana-fermion`) and multi-fermion contact vertices (`multi-fermion-contact`); Dirac–Weyl currents the tree lanes cannot close (`dirac-weyl-closure`); LC/NLC beyond the shipped N_c declaration (`colour-declaration-coverage`).
- **M-SM:** electroweak and mixed `[virt=QCD QED]` corrections (WP3.2, paused: EW native inputs EWI-N, the EW ladder and target EWI-7, EWI-9; photon-only QED on `loop_qcd_qed_sm` refused, `veto-unlabelled-counterterms`), the complex-mass scheme (EWI-10).
- **M-FINAL:** automatic rescue (none: no escalation by default; the rescue options left with the legacy engine), helicity and colour-flow selection at NLC and full colour (refused there; LC has it), SDK v4 extras, permutation aliases and process-group mappings, correlations composed with mappings, G1's provider protocol-mode columns, independent oracles of correlated Borns and virtuals (MQ-CORR).
- **Accepted at the pin** (coordinator 2026-10-29 ~11:25): A09@collinear (F7) — at the collinear points $\theta = 10^{-3}$–10^{-5} binary64 returns a typed `Unstable` and DD a typed `CutFailed{DegenerateGram}`, never a value (arbitrary precision is the user's route); A24 (F11) — rule 4 not formed at seven gluons (correlated trees beyond addendum H's memory), its poles equal NJet's DD.
- **Fast binary64 lanes:** NinjaRS light-like spinors near the $-z$ axis (~ 10^{-5}) and model units ~10^4 from GeV (silent percent-level errors with checks off; NinjaRS spinor package paused): the recommended checks flag them. Light masses at the fast thresholds now go to the generic engine (`onshell-threshold`).
- **No longer missing:** LC helicity and colour-flow selection (MQ-SELECT), the seven-gluon Ward check (MQ-WARD), a typed glibc error (`glibc-floor`, glibc ≥ 2.38). **Removed:** the legacy per-diagram engine (MQ-LEGACY-REMOVE; `loop.engine`, `--loop-engine` refused with the removal reason).

In development (HANDOFF §6): next milestone **M-SM** (electroweak and mixed corrections, complex-mass scheme; the paused `ew-native`, `ew-validate` and NinjaRS spinor packages resume there), then **M-FINAL** (the deferrals above, optimisation). Rust phase 1 (HANDOFF §0.4): D3 (plan lowering in Rust, binary plan member), A5′s tree program, C's seven-leg dipole metric tables. Prepared artifacts: the parameter program's SymJIT bytes still differ per compilation (values unaffected; upstream fix filed).

Performance gates and timing protocol: definitions

- **Gate.** Faster than MadLoop and at most 1.30× each matched Recola and OpenLoops time, on the **finite part**, from $2 \to 3$ upward, **taken at $B = 128$ only** (user 2026-10-28, MASTER_PLAN §5.8: $B = 1$ and the $2 \to 5$ rows' $B = 16$ are measured and reported with their ratios, without a verdict; a configuration without a $B = 128$ timing has none); $2 \to 2$ is reported without a gate. A process whose time per point at $B = 128$ is below 5 ms has no blocking gate: its ratio is shown, its verdict is "below 5 ms: not gated (user 2026-10-24)" (user decision; `data/milestone.json` `gate_decisions.threshold_rule`); optimisation effort goes to higher multiplicities. Aspiration: fastest of all tools from $2 \to 3$.
- **Protocol.** Providers in production mode with poles off where they allow it (OpenLoops `truepoles 0`, Recola without IR poles, MadLoop production build); pyAmpliCol finite-only (`poles = "none"`) with stability checks off, one pinned thread, binary64, 16 lanes, the same frozen points (pyAmpliCol receives their exact rationals, the providers their binary64 decimal text); since milestone m1025a pyAmpliCol's setup heap is frozen out of Python's cyclic collector before the warm-up (a loaded runtime holds no generation objects; in the committed harness since `de8d3eb8`, which times G1 at `e18a024b`; Section 3). Statistic: median per point; the uncertainty band combines pyAmpliCol's block spread and the provider's repeat spread; a gate whose band contains the limit is **open**.
- **Timing boundary.** pyAmpliCol: the public v2 runtime call (`NativeExecutor.evaluate`) for a batch of B distinct points, everything included (per-point inputs, loop stage, assembly); the public `Runtime.evaluate` of a loaded artifact adds its input parsing (Section 3). Providers have no batch API: one time per point serves every B. Generation: pyAmpliCol `LoopSlice.generate` plus the plan export, in G1 on one pinned core, cold (fresh cache directory) since `11542fd5` (m1026a); a development record's generation in Table 1 ran on the shared host under load (threads per increment: Table 14), without a multiplier in a G1 row; MadLoop graph generation, export and compilation; Recola process generation; OpenLoops libraries are prebuilt (no generation time).
- **Labels.** Table 1 and Figure 1 show the most recent measured configuration of each cell (main-configuration rule of `scripts/build_summary.py`; `data/summary.json` names the configuration and source of every value); the column header of Table 1 states it once, with branch and commit, and "Reading the summary" says when it is a development measurement not yet on `loop`. The detailed G1 tables (Section 3) show the configuration on `loop` and, labelled, its history: the runs of milestones m1028a (`28a16255`), m1026a (`11542fd5`) and m1025a (`1d27426b`) and of G-int (`4e9b5a9e`), the Python per-point inputs before G-int and the development runs of `39824f9b`. Missing values are marked, never estimated or extrapolated; test durations are never used as event timings. The legacy per-diagram engine's timings of 18 October are history (data kept in `data/measurements.json`, not shown).

2 QCD capability matrix (M-QCD)

Table 2: **QCD capability matrix** at e18a024b, per class of m-qcd/PLAN.md §1.1 (campaign of 2026-10-29, data/mqcd.json; legend below the table, rows: Table 21).

Class	Rows	Public route	DD vs ref.	binary64	Catani (DD)	LC/NLC	FFT	Status
Q-massless gluons and massless quark lines, every initial state · up to $2 \to 5$	**22** pass 20 partial 1 fail 1	A12 ✓ DD 3.8e−14 · 3 pts	**1.2e−11** A18	**1.1e−10** A18 · ref. 7.3e−7†	**1.2e−22** A09 · A24 not formed	rule 7 ✓ 8 rows · smoke pass	**×0.46-5.92** seed · 9 rows; total ×0.93–1.29	accepted 20 of 22 pass; F7 A09@collinear (accepted); F11 A24 (accepted)
Q-top $t\bar{t}$ + jets, $t\bar{t}$ with quark lines, four tops · up to $2 \to 4$	**6** pass 6	M01@P0, M01@P2 ✓ DD 2.0e−15 · 8 pts	**3.4e−13** B03	**1.2e−10** B01 · ref. 1.7e−7†	**3.0e−26** B01	rule 7 ✓ 1 row · smoke pass	**×1.13** seed · 1 row; total ×0.96	pass
Q-heavy massive b (4FS) and c, b-Yukawa Higgs · up to $2 \to 4$	**10** pass 10	C01 ✓ DD 9.3e−16 · 4 pts	**5.0e−14** C02	**2.3e−10** C05 · ref. 2.3e−10	**1.1e−28** C09	rule 7 ✓ 1 row · smoke pass	auto no cost row	pass
Q-EWBorn QCD corrections to Borns with QED >= 1 · up to $2 \to 5$	**25** pass 25	D05 ✓ DD 2.3e−14 · 4 pts	**1.8e−12** D28	**1.4e−11** D28 · ref. 3.1e−11†	**4.5e−28** D29 · D19 no sector	rule 7 ✓ 1 row · smoke pass	auto no cost row	pass
Q-ttX $t\bar{t}$ + H/Z, $t\bar{t}$ + leptons, e+ e- > t t~ (h) · up to $2 \to 5$	**4** pass 4	E01 ✓ DD 3.6e−13 · 4 pts	**3.6e−13** E01	**7.4e−12** E01 · ref. 5.7e−9†	**5.7e−27** E06	rule 7 ✓ 1 row · smoke pass	auto no cost row	pass
Q-singletop s- and t-channel, tW, 4FS t-channel · up to $2 \to 3$	**5** pass 5	F03 ✓ DD 1.7e−15 · 4 pts	**8.0e−15** C07	**1.6e−14** C07 · ref. 1.6e−14	**3.4e−30** F04 · C07 no sector	rule 7 ✓ 1 row · smoke pass	auto no cost row	pass
Q-mixed several Born orders, every squared split order · up to $2 \to 2$	**2** pass 2	G02 ✓ DD 2.0e−15 · 4 pts	**4.6e−15** G01	**3.1e−14** G02 · ref. 2.9e−14	**3.7e−30** G02	smoke pass no rule-7 row	auto no cost row	pass
Q-CKM non-diagonal CKM on W lines · up to $2 \to 2$	**3** pass 3	H01 ✓ DD 1.4e−15 · 4 pts	**1.4e−15** H01	**7.2e−14** H01 · ref. 7.3e−14	**4.6e−29** H01	smoke pass no rule-7 row	auto no cost row	pass
Q-multi multiparticle labels (p p > t t~) · up to $2 \to 2$	**1** pass 1	M01@P0, M01@P2 ✓ DD 2.0e−15 · 8 pts	**2.0e−15** M01	**4.1e−14** M01 · ref. 4.3e−14	**4.0e−30** M01	smoke pass no rule-7 row	auto no cost row	pass

Rows: campaign rows and their status (passing every rule of their tier; partial: a rule not formed); class status **accepted**: every other row is an open finding the coordinator accepted for the gate. **Public route**: the class representative generated by Generator on the shipped loop_sm, Runtime.load and evaluate, against MadLoop (rules 1–3; Q-top and Q-multi by the campaign's public variants of $pp \to t\bar{t}$). **DD vs ref.**: largest relative deviation of the double-double Laurent coefficients (Born, finite, poles, every sector) from the binding reference (rule 1; MadLoop in quadruple precision unless named). **binary64**: from DD where binary64 is asserted (rule 2, estimate $E \le 10^{-9}$); "ref." from the reference at every point (information; † the class has points with $E > 10^{-9}$, where DD is asserted). **Catani**: DD poles against the Catani operator (rule 4, every point and predicted sector). **LC/NLC**: rule 7 (formal-N_c projection = N_c scan, and MQ-NC's independent oracles) and the smoke on every shipped restriction. **FFT**: seed contraction FFT ÷ direct of the class's cost rows (Table 3); auto runs the certified FFT wherever the plan has a certified pair.

Reading the matrix. The campaign ran every row of its tier at the pin as whole test files, one process each, the manifest's points in binary64 and double-double (rules 1–4; rules 5–7 and the extras on the rows that name them; m-qcd/PLAN.md §1.4 under the user's ruling of 2026-10-29 ~10:20: a deviation below ~10^{-5} is not a finding). Its counts: 76 pass, 1 partial, 1 fail, 0 pending, 0 not run. The two non-passing rows are accepted limitations, not wrong values: A24 ($gg \to ggggg$, F11) forms no rule 4 because the seven-gluon colour-correlated trees exceed addendum H's memory budget, and its poles equal NJet's double-double values (rule 1); A09 ($u\bar{u} \to ggg$, F7) fails only through its collinear variant, where the executor returns typed failures and no value (binary64 Unstable at $\theta = 10^{-3}$–10^{-5}, DD CutFailed{DegenerateGram} at 10^{-4}, 10^{-5}); A09 itself and its soft variant pass. D19 and C07 have no predicted pole sector (MQ-EWJOIN's separated-exchange rule: their QCD loop selection omits IR-divergent loops, as MadLoop's does) and are checked against MadLoop per split order. The public-route representatives ran on a scratch copy of the manifest (G02 with explicit Born limits, H01 without the tools route's CKM twin relation); MQ-MODEL's public-equals-tools-view tests passed at the pin.

Table 3: **FFT seed contraction against the direct one** (MQ-FFT, color.contraction): time for the batch of 128 G1-stream points (binary64, poles none, checks off, one pinned thread; g g > u u~ d d~ on the six-gluon stream; one paired round, best of two below $2 \to 5$; indicative), FFT / direct and their ratio (green faster, red slower; Table 1's scale), for the whole evaluation and for the executor's seed-contraction stage. The values agree to rounding at stable points; at the cost runs' worst binary64 points (21 point/carrier checks, binary64-unstable: E up to $O(10^2)$) FFT − direct stays within 2.8 % of the acceptance envelope $10(E_{\mathrm{FFT}} + E_{\mathrm{direct}}) + 16uS$ in binary64 and within 1.7 % in DD (relative ≤ 1.1×10⁻¹⁹); test_loop_v2_seed_fft.py (default, nightly, seven legs) passes at the pin.

Process	Class	group	pairs	evaluation, s	ratio	seed contraction, ms	ratio
$gg \to gg$	Q-massless	S_3	2	0.266 / 0.205	**×1.29**	2.00 / 0.339	**×5.92**
$gg \to ggg$	Q-massless	S_4	2	2.58 / 2.76	**×0.93**	10.5 / 7.03	**×1.50**
$gg \to gggg$	Q-massless	S_5	2	49.3 / 52.1	**×0.95**	180 / 255	**×0.70**
$gg \to ggggg$	Q-massless	S_6	2	1275 / 1246	**×1.02**	8 160 / 17 700	**×0.46**
$u\bar{u} \to ggg$	Q-massless	S_3	19	0.800 / 0.825	**×0.97**	11.4 / 6.73	**×1.69**
$u\bar{u} \to gggg$	Q-massless	S_4	40	10.6 / 10.2	**×1.04**	357 / 312	**×1.15**
$u\bar{u} \to ggggg$	Q-massless	S_5	74	240 / 247	**×0.97**	24 300 / 34 700	**×0.70**
$gg \to t\bar{t}gg$	Q-top	S_4	40	13.2 / 13.7	**×0.96**	224 / 198	**×1.13**
$gg \to u\bar{u}d\bar{d}$	Q-massless	S_2	2340	4.95 / 4.55	**×1.09**	260 / 89.8	**×2.89**
$u\bar{u} \to d\bar{d}g$	Q-massless · no certified pair: direct	—	—	0.519 / 0.420	**×1.24**	2.88 / 2.88	**×1.00**

3 Gate G1: end-to-end timings of the v2 engine

G1 times the complete Born–virtual evaluation, **finite part only**, through the call of the public v2 runtime (`NativeExecutor.evaluate`, which `V2VirtualRuntime` calls) with stability checks off, on one pinned core and one thread, binary64, 16 SIMD lanes, in batches of B distinct points of a frozen stream (REQUIREMENTS_ADDENDUM D, plan §5.8); pyAmpliCol receives the stream's exact rational points, the providers their binary64 decimal text. The configurations:

- **native inputs, on `loop`** — the engine at `e18a024b` (milestone M-QCD: the engine of m1028a, `28a16255`, with the 15 packages of Section 1; build `loop-native-v2-slice-d0fef168`, digest `1be09f92`): the executor forms the trees, the Born and the counterterm inputs per point (binary64 SymJIT plane applications, tree closures in lanes of points), skips the counterterm interactions that only feed poles in finite-only mode, then runs the loop stage. $gg \to ggggg$ is timed from its stored process. The public path ($u\bar{u} \to d\bar{d}g$): the public generator's artifact has the harness's plan digest, `Runtime.load` in a fresh interpreter makes no generation call, and the loaded executor returns the timed run's binary64 values bitwise at all 128 stream points (`public_path_check` in `data/g1.json`). **Values against m1028a** (`values_note`): bitwise in 19 of 30 configurations (every leading-colour one and the quark-only full-colour ones); the full colour of the rows with gluons moved at rounding level at stable points (Borns bitwise); every change above 10^{-5} lies at a binary64-unstable point, at most 0.11 of its estimate E (stability rows added there). At 6 of $gg \to ggggg$'s 128 stream points (17, 89, 93, 95, 109, 112) binary64 is wrong by $O(1)$ or more in both engines; the recommended checks flag every one (E = 8 to 5×10^3), the timing runs evaluate them with checks off. Earlier campaigns (m1028a `28a16255`, m1026a `11542fd5`, m1025a `1d27426b`, G-int `4e9b5a9e`, the development campaign of `39824f9b`) are `earlier_runs`, history;
- **Python inputs** history — the engine on `loop` before G-int, G1 at `1c971289` (LC at `d591a2f7`): trees, Born amplitudes and counterterm scalars formed per point in Python, then the loop stage natively.

Timing protocol at `e18a024b` (`timing_protocol` in `data/g1.json`): the committed harness, binary64, finite part only, no stability check, one thread on one pinned core, setup heap frozen out of Python's cyclic collector before the warm-up, the frozen streams. The native digest changed (f813c625 ⟶ `1be09f92`), so every configuration was re-timed at $B = 128$ (three blocks; at $2 \to 5$ after a 16-point warm-up), full and leading colour, in one round (timings indicative, user 2026-10-28: no repeat for host contention; no verdict was open, so no verdict repeat). $B = 1$ and the $2 \to 5$ rows' $B = 16$ were **not re-timed**: the tables show m1028a's runs (`28a16255`, each configuration's last earlier run) marked °. Host: AMD EPYC 9754, shared host, cores 128-223 without SMT siblings: milestone m1029-mqcd's G1 on 2026-10-29 (17:21-19:07 UTC), at most one job per CCX of 8 cores, each on the least busy core of a CCX whose least busy core was below 50 % busy (cores per record: host_cpus); host load average 52-247 (other users' processes and the milestone's whole-suite gate ran at the same time); timings indicative (user 2026-10-28), one round, no repeats for contention.

Generation (`generation_protocol`): every row generated cold in its own process with fresh caches, as in m1026a; against m1028a, model + generate + export changed by −6 % to +19 % (median −3 %), plan sizes equal everywhere, and the seven-gluon plans are byte-identical ($gg \to ggggg$: full colour 170 + 214 s, 9.5 GiB; LC 173 + 207 s, 10.0 GiB). The provider records are reused unchanged (`provider_records`).

Table 4: **Gate G1, full colour**, ms per point (gate statistic: median), finite part, one thread. Rows $B = 1$ and $B = 128$ (the Python-input $2 \to 4$ rows only at $B = 1$: their Python inputs took seconds per point). Multiplier and colour: time ÷ the fastest provider of the row (legend of Table 1). Grey rows: generation — pyAmpliCol generation plus plan export with its peak memory; MadLoop graph generation, export and compilation · the share of its loop-numerator coefficients in its time (Table 6); Recola process generation; OpenLoops: the prebuilt library and its load plus first call. MadLoop $u\bar{u} \to gggg$ was not exported: the $gg \to gggg$ export took 2.4 h here (HANDOFF §0.1), and OpenLoops binds the gate. Gate: upper line Python inputs (history), lower line native inputs (`loop`); the number is the time over the binding limit (MadLoop's time or 1.30× the faster of Recola and OpenLoops); **open**: the band contains the limit; **not gated**: the configuration runs below 5 ms per point at $B = 128$ (user rule of 2026-10-24; ratio as measured); **reported**: the verdict is taken at $B = 128$ only, so the $B = 1$ ratios are measured and shown without a verdict (user 2026-10-28; bands: Table 5); ° native $B = 1$: the run of `28a16255`, not re-timed at `e18a024b` (uncoloured, history; its ratio over this campaign's limit).

Process	B	Python inputs[hist]		Native inputs, loop		MadLoop		Recola		OpenLoops		Gate
$gg \to gg$ 2 → 2, no gate	1	**140**	**×69**	15.0°	×7.4	**2.02**	fastest	**10.0**	**×5.0**	**3.20**	**×1.6**	no gate
	128	**132**	**×66**	**1.47**	**×0.73**	**2.02**	fastest	**10.0**	**×5.0**	**3.20**	**×1.6**	no gate
	gen.	10.4 s · 0.3 GiB		8.21 s · 0.6 GiB		9.70 s · 64 %		0.142 s		ppjj · 6.80 ms		
$u\bar{u} \to d\bar{d}g$ 2 → 3	1	**40.6**	**×27**	10.5°	×7.0	**3.17**	**×2.1**	**3.18**	**×2.1**	**1.52**	fastest	reported 20.61 reported 5.35°
	128	**32.4**	**×21**	**2.20**	**×1.5**	**3.17**	**×2.1**	**3.18**	**×2.1**	**1.52**	fastest	**fail** ×16 **not gated** 1.12
	gen.	9.94 s · 0.3 GiB		9.28 s · 0.6 GiB		9.87 s · 73 %		0.406 s		ppjjj · 4.73 ms		

Process	B	Python inputs[hist]	Native inputs, `loop`	MadLoop	Recola	OpenLoops	Gate
$u\bar{u} \to u\bar{u}g$ $2 \to 3$	1	**68.9** ×24	18.5° ×6.3	**8.59** ×2.9	**5.55** ×1.9	**2.92** fastest	reported 18.12 reported 4.87°
	128	**54.5** ×19	**2.60** ×0.89	**8.59** ×2.9	**5.55** ×1.9	**2.92** fastest	**fail** ×14 **not gated** 0.68
	gen.	10.4 s · 0.3 GiB	8.64 s · 0.6 GiB	10.9 s · 75 %	0.462 s	ppjjj · 8.43 ms	
$u\bar{u} \to e^-e^+g$ $2 \to 3$	1	*not generable*	4.37° ×4.6	**0.948** fastest	**1.08** ×1.1	*no library*	— reported 4.61°
	128	*not generable*	**0.691** ×0.73	**0.948** fastest	**1.08** ×1.1	*no library*	— **not gated** 0.73
	gen.	CT coverage refused	12.3 s · 0.6 GiB	14.0 s · 79 %	0.701 s	—	
$u\bar{u} \to ggg$ $2 \to 3$	1	**215** ×17	48.1° ×3.9	**28.2** ×2.3	**25.3** ×2.0	**12.4** fastest	reported 13.38 reported 2.98°
	128	**179** ×14	**5.26** ×0.42	**28.2** ×2.3	**25.3** ×2.0	**12.4** fastest	**fail** ×11 **pass** 0.33
	gen.	12.3 s · 0.4 GiB	9.28 s · 0.6 GiB	17.6 s · 76 %	0.634 s	ppjjj · 23.7 ms	
$gg \to ggg$ $2 \to 3$	1	**2 760** ×28	205° ×2.1	**247** ×2.5	**351** ×3.6	**97.0** fastest	reported 21.86 reported 1.62°
	128	**2 560** ×26	**19.5** ×0.20	**247** ×2.5	**351** ×3.6	**97.0** fastest	**fail** ×20 **pass** 0.15
	gen.	27.9 s · 0.6 GiB	10.2 s · 0.6 GiB	47.0 s · 78 %	6.63 s	ppjjj · 159 ms	
$u\bar{u} \to e^-e^+gg$ $2 \to 4$	1	*not generable*	44.8° ×4.4	**27.3** ×2.7	**10.2** fastest	*no library*	— reported 3.37°
	128	*not generable*	**4.93** ×0.48	**27.3** ×2.7	**10.2** fastest	*no library*	— **not gated** 0.37
	gen.	CT coverage refused	13.1 s · 0.6 GiB	24.8 s · 83 %	0.833 s	—	
$gg \to gggg$ $2 \to 4$	1	**101 000** ×36	3 850° ×1.4	**38 700** ×14	**17 600** ×6.2	**2 840** fastest	reported 27.42 reported 1.04°
	128	*not timed*	**384** ×0.14	**38 700** ×14	**17 600** ×6.2	**2 840** fastest	— **pass** 0.10
	gen.	335 s · 4.1 GiB	31.9 s · 1.3 GiB	8 680 s · 51 %	2 130 s	ppjjjj · 4 950 ms	
$gg \to t\bar{t}gg$ $2 \to 4$	1	**7 310** ×15	817° ×1.7	**5 450** ×12	**1 970** ×4.2	**472** fastest	reported 11.92 reported 1.33°
	128	*not timed*	**92.2** ×0.20	**5 450** ×12	**1 970** ×4.2	**472** fastest	— **pass** 0.15
	gen.	62.5 s · 0.9 GiB	14.4 s · 0.7 GiB	218 s · 64 %	61.4 s	pptttj · 821 ms	
$u\bar{u} \to gggg$ $2 \to 4$	1	**5 760** ×15	801° ×2.1	*not timed*	**908** ×2.4	**374** fastest	reported 11.85 reported 1.65°
	128	*not timed*	**73.4** ×0.20	*not timed*	**908** ×2.4	**374** fastest	— **pass** 0.15
	gen.	56.1 s · 0.7 GiB	14.4 s · 0.7 GiB	—	32.8 s	ppjjjj · 606 ms	
$u\bar{u} \to e^-e^+d\bar{d}g$ $2 \to 5$	1	*not timed*	*not timed*	**132** ×3.7	**35.3** fastest	**51.0** ×1.4	— —
	128	*not timed*	**16.8** ×0.48	**132** ×3.7	**35.3** fastest	**51.0** ×1.4	— **pass** 0.37
	gen.	—	17.1 s · 0.7 GiB	58.6 s	6.73 s	ppljjj · 108 ms	
$u\bar{u} \to e^-e^+ggg$ $2 \to 5$	1	*not timed*	*not timed*	**1 420** ×5.7	**251** fastest	**346** ×1.4	— —
	128	*not timed*	**58.1** ×0.23	**1 420** ×5.7	**251** fastest	**346** ×1.4	— **pass** 0.18
	gen.	—	17.1 s · 0.7 GiB	146 s	13.4 s	ppljjj · 727 ms	
$gg \to t\bar{t}e^-e^+g$ $2 \to 5$	1	*not timed*	*not timed*	**3 040** ×7.5	**620** ×1.5	**404** fastest	— —
	128	*not timed*	**63.1** ×0.16	**3 040** ×7.5	**620** ×1.5	**404** fastest	— **pass** 0.12
	gen.	—	17.9 s · 0.8 GiB	165 s	27.7 s	pplltj · 1 060 ms	
$u\bar{u} \to ggggg$ $2 \to 5$	1	*not timed*	*not timed*	**569 000** ≥×3.1	**≤ 182 000** (outside protocol, poles on: upper bound)	*no library*	— —
	128	*not timed*	**1 750** ≥×0.0096	**569 000** ≥×3.1	**≤ 182 000** (outside protocol, poles on: upper bound)	*no library*	— **pass** 0.0031
	gen.	—	138 s · 3.0 GiB	44 000 s	—	—	
$gg \to ggggg$ $2 \to 5$, no gate (no matched provider)	1	*not timed*	*not timed*	*not exported*	*not generated*	*no library*	no gate
	128	*not timed*	**7 040**	*not exported*	*not generated*	*no library*	no gate
	gen.	—	383 s · 9.5 GiB	—	—	—	

The providers run in production mode on the same stream: OpenLoops with `truepoles 0`, Recola without IR poles (one amplitude call per point), MadLoop at its production stability setting on exports built with MG5's production flags. OpenLoops has no installed library with QCD corrections to $\ell^+\ell^-$ + 1 or 2 jets (`ppllj_ew` carries electroweak corrections only; existing provider builds only), so the V+jets gates bind on MadLoop and Recola. Providers have no batch API: one time per point serves every B. **Gate statistic:** the median per point on both sides — over pyAmpliCol's

timed blocks of one B, and over the provider's timed points (the stream's first 16; 8 for MadLoop $gg \to gggg$, Recola $gg \to gggg$, OpenLoops $gg \to gggg$, Recola $u\bar{u} \to gggg$, MadLoop $u\bar{u} \to ggggg$ and Recola $u\bar{u} \to ggggg$). The provider times are the records of the measure-g1 campaign (23–24 October, on cores 129–345), not re-run since, and for the $2 \to 5$ rows the production-mode runs of package `measure-g1-7leg` (25 October, each provider with its production Collier cache: on for Recola's COLLIER 1.2.8 and OpenLoops, off for MadLoop's COLLIER 1.2.9 with its seven-leg cache defect; $u\bar{u} \to ggggg$ imported from the reference campaign's three points); pyAmpliCol ran on 26 cores between 135 and 216 (per run: `host_cpus` in `data/g1.json`) on 2026-10-29, at most one job per eight-core group sharing an L3 cache, each on the least busy core of a group whose least busy core was below 50 % busy, while the milestone's whole-suite gate and other users' processes loaded the host (`host` in `data/g1.json`: host load average 52-247 (other users' processes and the milestone's whole-suite gate ran at the same time)). The three $B = 128$ blocks of a process spread by at most 5.7 % (half the range over the median; $u\bar{u} \to d\bar{d}g$ 2.164, 2.415, 2.200 ms). The gate's uncertainty is pyAmpliCol's block spread (half the range of three blocks, the median's standard error from five) plus the provider's half-range over its repeated runs (8 %, the campaign's largest, for the single runs of Recola $gg \to gggg$, Recola $u\bar{u} \to gggg$, OpenLoops $u\bar{u} \to e^-e^+d\bar{d}g$, MadLoop $u\bar{u} \to e^-e^+d\bar{d}g$, OpenLoops $u\bar{u} \to e^-e^+ggg$, Recola $u\bar{u} \to e^-e^+ggg$, MadLoop $u\bar{u} \to e^-e^+ggg$, OpenLoops $gg \to t\bar{t}e^-e^+g$, Recola $gg \to t\bar{t}e^-e^+g$, MadLoop $gg \to t\bar{t}e^-e^+g$ and MadLoop $u\bar{u} \to ggggg$). A gate passes or fails only if the whole band does; otherwise it is **open** and the configuration is repeated once (plan §5.8 item 10). Every timed call returned all its points (failed points: 0); rescue is off by protocol (no checks, no escalation), so the rescue rate is 0.

Table 5: **The gate at every batch size** (native inputs on `loop`, full colour): time over the binding limit and its uncertainty band. **The speed gate is taken at $B = 128$ only** (user 2026-10-28, MASTER_PLAN §5.8; `data/milestone.json` `gate_decisions.gate_batch`): only that column has a verdict and a colour (green pass, red fail, amber open, grey not gated: below 5 ms per point at $B = 128$, user rule of 2026-10-24); every other batch size is **reported**: measured, its ratio and band shown in a white cell, no verdict. $B = 1$ is timed with the default 16 lanes (a one-point call pays a 16-lane chunk); the one-lane column is the development campaign's (`39824f9b`, history: the public path always runs 16 lanes in binary64), a configuration without a $B = 128$ timing and therefore without any verdict (gate missing: no B = 128 timing (timed at B = 1); G1 is taken at B = 128 only (user 2026-10-28)). This campaign timed $B = 128$ only (no verdict was open, so no repeat: `repeats` in `data/g1.json`); ° grey: the earlier run of `28a16255`, `39824f9b`, `4e9b5a9e` at that batch size, its median over the same binding limit with its time (no band; an earlier engine under the committed protocol; history).

Process	$B = 1$, 16 lanes		$B = 1$, 1 lane (39824f9b)		$B = 8$		$B = 16$		$B = 32$		$B = 128$ (gate)	
$u\bar{u} \to d\bar{d}g$	*5.35°*	10.5 ms	2.13	2.10–2.16	*1.46°*	2.87 ms	—		*1.15°*	2.26 ms	**1.12**	1.04–1.19
$u\bar{u} \to u\bar{u}g$	*4.87°*	18.5 ms	1.63	1.59–1.67	*1.16°*	4.40 ms	—		*0.87°*	3.31 ms	**0.68**	0.67–0.70
$u\bar{u} \to e^-e^+g$	*4.61°*	4.37 ms	1.61	1.60–1.62	*1.07°*	1.01 ms	—		*0.85°*	0.805 ms	**0.73**	0.73–0.73
$u\bar{u} \to ggg$	*2.98°*	48.1 ms	0.84	0.81–0.86	*0.68°*	11.0 ms	—		*0.74°*	11.9 ms	**0.33**	0.32–0.34
$gg \to ggg$	*1.62°*	205 ms	0.45	0.45–0.45	*0.42°*	52.7 ms	—		*0.34°*	42.9 ms	**0.15**	0.15–0.15
$u\bar{u} \to e^-e^+gg$	*3.37°*	44.8 ms	0.78	0.78–0.78	*0.62°*	8.20 ms	—		*0.42°*	5.54 ms	**0.37**	0.36–0.38
$gg \to gggg$	*1.04°*	3 850 ms	0.31	0.30–0.31	*0.30°*	1 120 ms	—		*0.26°*	977 ms	**0.10**	0.10–0.11
$gg \to t\bar{t}gg$	*1.33°*	817 ms	0.37	0.36–0.37	*0.35°*	215 ms	—		*0.28°*	172 ms	**0.15**	0.15–0.15
$u\bar{u} \to gggg$	*1.65°*	801 ms	0.43	0.43–0.44	*0.47°*	230 ms	—		*0.37°*	178 ms	**0.15**	0.15–0.15
$u\bar{u} \to e^-e^+d\bar{d}g$	—		—		—		*0.34°*	15.4 ms	—		**0.37**	0.33–0.40
$u\bar{u} \to e^-e^+ggg$	—		—		—		*0.18°*	57.2 ms	—		**0.18**	0.16–0.19
$gg \to t\bar{t}e^-e^+g$	—		—		—		*0.12°*	61.2 ms	—		**0.12**	0.11–0.13
$u\bar{u} \to ggggg$	—		—		—		*0.0030°*	1 730 ms	—		**0.0031**	0.0028–0.0033

3.1 Where the time goes

Table 6: **Stage split**, ms per point, full colour, at the largest batch timed (Python-input $2 \to 4$: $B = 1$). Trees/Born/CT: the per-point inputs, timed in Python before G-int (history: trees, counterterm program, Born and counterterm colour projections, scalars); on `loop` the executor forms them and, since lc-perf (`88657260`), times them itself (its `trees` stage: programs, closures, interferences). Numerator: seeds, fused factors and the dressing of the open-loop objects (the pure numerator evaluation). Scatter: closures into NinjaRS lanes. NinjaRS: the reduction without the masters; masters: the master integrals (canonical arguments, cache, oneloopmaster). Glue: point parameters, carrier conversion, assembly and, where the inputs are timed, the rest of the Python call (all rows of this table). MadLoop (instrumented copy; its points less one, 3 for $gg \to gggg$): Born/counterterm HELAS calls, loop-numerator coefficients, reduction plus masters, residual (stability test, colour sums). Bar: shares, colours as the column heads.

Process		Trees, Born, CT	Numerator	Scatter	NinjaRS	Masters	Glue	Total	Shares
$gg \to gg$	native, `loop`	0.188	0.829	0.0701	0.115	0.0369	0.239	**1.47**	
	Python^{hist}	128	0.851	0.0919	0.119	0.0381	2.96	**132**	
	MadLoop	0.0315	1.27	—	0.585		0.107	**1.99**	
$u\bar{u} \to d\bar{d}g$	native, `loop`	0.113	0.878	0.421	0.192	0.163	0.427	**2.20**	
	Python^{hist}	28.1	0.749	0.462	0.137	0.111	2.66	**32.4**	
	MadLoop	0.0479	2.35	—	0.683		0.133	**3.22**	

Process		Trees, Born, CT	Numerator	Scatter	NinjaRS	Masters	Glue	Total	Shares
$u\bar{u} \to u\bar{u}g$	native, `loop`	0.110	1.21	0.413	0.231	0.164	0.473	**2.60**	
	Python^{hist}	48.5	1.23	0.479	0.232	0.171	3.99	**54.5**	
	MadLoop	0.137	6.44	—	1.15		0.814	**8.54**	
$u\bar{u} \to e^-e^+g$	native, `loop`	0.0436	0.268	0.124	0.0297	0.0184	0.207	**0.691**	
	MadLoop	0.0184	0.754	—	0.133		0.0460	**0.951**	
$u\bar{u} \to ggg$	native, `loop`	0.391	3.03	0.446	0.494	0.237	0.660	**5.26**	
	Python^{hist}	168	2.92	0.476	0.459	0.225	5.75	**179**	
	MadLoop	0.346	18.5	—	2.54		2.87	**24.3**	
$gg \to ggg$	native, `loop`	4.03	11.5	0.486	2.02	0.488	0.952	**19.5**	
	Python^{hist}	2 530	11.7	0.519	2.03	0.483	10.5	**2 560**	
	MadLoop	2.42	184	—	11.3		39.5	**237**	
$u\bar{u} \to e^-e^+gg$	native, `loop`	0.172	2.84	0.621	0.451	0.253	0.594	**4.93**	
	MadLoop	0.289	22.7	—	2.38		2.11	**27.5**	
$gg \to gggg$	native, `loop`	101	216	2.38	45.8	5.85	11.7	**384**	
	Python^{hist}	97 300	3 750	4.08	38.6	5.81	137	**101 000**	
	MadLoop	88.9	18 600	—	209		17 700	**36 500**	
$gg \to t\bar{t}gg$	native, `loop`	11.0	45.4	0.937	20.9	6.30	7.93	**92.2**	
	Python^{hist}	6 490	751	3.55	20.7	6.58	29.3	**7 310**	
	MadLoop	24.2	3 290	—	105		1 740	**5 160**	
$u\bar{u} \to gggg$	native, `loop`	7.55	47.3	1.73	9.06	2.89	4.51	**73.4**	
	Python^{hist}	4 910	806	2.01	9.64	2.99	24.1	**5 760**	
$u\bar{u} \to e^-e^+d\bar{d}g$	native, `loop`	0.644	9.02	1.93	2.04	1.32	1.86	**16.8**	
$u\bar{u} \to e^-e^+ggg$	native, `loop`	1.49	40.2	2.13	8.20	2.75	3.45	**58.1**	
$gg \to t\bar{t}e^-e^+g$	native, `loop`	2.95	39.7	0.835	11.0	4.13	4.68	**63.1**	
$u\bar{u} \to ggggg$	native, `loop`	226	1 140	9.86	265	45.9	55.8	**1 750**	
$gg \to ggggg$	native, `loop`	2 890	3 620	11.3	443	40.9	28.8	**7 040**	

Verdict of gate G1 (processes from $2 \to 3$; $2 \to 2$ without a gate).

- **Native inputs on `loop`** (G1 at `e18a024b`) at $B = 128$: **pass** for all 9 gated processes, $u\bar{u} \to ggg$, $gg \to ggg$, $gg \to gggg$, $gg \to t\bar{t}gg$, $u\bar{u} \to gggg$, $u\bar{u} \to e^-e^+d\bar{d}g$, $u\bar{u} \to e^-e^+ggg$, $gg \to t\bar{t}e^-e^+g$ and $u\bar{u} \to ggggg$. **Not gated** (below 5 ms per point, user decision of 2026-10-24; ratios as measured): $u\bar{u} \to d\bar{d}g$ 2.20 ms (1.12 of its limit), $u\bar{u} \to u\bar{u}g$ 2.60 ms (0.68 of its limit), $u\bar{u} \to e^-e^+g$ 0.691 ms (0.73 of its limit) and $u\bar{u} \to e^-e^+gg$ 4.93 ms (0.37 of its limit). $B = 1$ (16 lanes) and the $2 \to 5$ rows' $B = 16$ are **reported** without a verdict (user 2026-10-28): $B = 1$ at ×1.0–5.3 of the limit (a one-point call pays a 16-lane chunk, Table 5); with one lane per call, in the development runs of `39824f9b` (history), ×0.31–2.1.
- **Python inputs** history (the engine before G-int, G1 at `1c971289`): failed every process it generated, at $B = 128$ at ×11–20 of the binding limit, the per-point Python inputs taking 87 %–99 % of its time (Table 6); at $B = 1$ at ×12–27, where $gg \to gggg$, $gg \to t\bar{t}gg$ and $u\bar{u} \to gggg$ were timed only. It did not generate $u\bar{u} \to e^-e^+g$ and $u\bar{u} \to e^-e^+gg$ (Furry zeros, WP3.5; Z fermion signs, WP3.2).
- **Fastest of all tools** (aspiration), native inputs at $B = 128$, within the uncertainty band: faster than every provider for $u\bar{u} \to u\bar{u}g$, $u\bar{u} \to e^-e^+g$, $u\bar{u} \to ggg$, $gg \to ggg$, $u\bar{u} \to e^-e^+gg$, $gg \to gggg$, $gg \to t\bar{t}gg$, $u\bar{u} \to gggg$, $u\bar{u} \to e^-e^+d\bar{d}g$, $u\bar{u} \to e^-e^+ggg$ and $gg \to t\bar{t}e^-e^+g$; faster than every provider with a production-mode time for $u\bar{u} \to ggggg$ (MadLoop; Recola has only an upper bound of its production time, ≤ 182 s); at parity with the fastest for none; slower than the fastest for $u\bar{u} \to d\bar{d}g$ (OpenLoops). NJet, not a matched provider, takes 4.59 s (NJet 3.1.2) for its two-evaluation full-colour $gg \to gggg$ test (Table 15).
- Seven-leg rows ($u\bar{u} \to e^-e^+d\bar{d}g$, $u\bar{u} \to e^-e^+ggg$, $gg \to t\bar{t}e^-e^+g$, $u\bar{u} \to ggggg$, $gg \to ggggg$) are generated by the WP3.1 generator, on `loop` since G-int, and timed by G1 since package `measure-g1-7leg` (full and leading colour at $B = 16$ and 128; re-timed at `e18a024b`); $gg \to ggggg$ has no matched provider timing (addendum F, H): reported without a gate.

Table 7: **Levers**, measured on the gated rows of Table 6 ($2 \to 3$ and beyond; shares at $B = 128$). Gains are measured where a column says so; otherwise they are the stage's share, an upper bound.

Lever	Evidence	Expected gain	Owner
Native per-point inputs (WP3.1 I3–I6: trees, Born, counterterm scalars in the executor)	Python inputs 87 %–99 % of the Python-input time at $B = 128$ (history)	measured, Python ÷ native inputs at equal B: — at $B = 1$, ×15–131 at $B = 128$ ($2 \to 3$)	done: G-int, on loop since 4e9b5a9e
Binary64 point inputs (the providers' input; the protocol passes the stream's exact rationals)	paired run on the loaded artifact of G-int (4e9b5a9e, history), $B = 128$: $u\bar{u} \to d\bar{d}g$ 1.68 against 2.29 ms, $u\bar{u} \to u\bar{u}g$ 2.94 against 3.39 ms	measured: ×1.37 and ×1.15 (information; not a G1 configuration)	none: moot under the 5 ms gate rule (user, 2026-10-24); the protocol keeps exact inputs
Compiled native trees (kernels as the tree engine's SymJIT planes, lanes batched)	native trees (with the Born and counterterm inputs; the executor's timer) 3 %–26 % of the time at $B = 128$; $gg \to gggg$ 26 %	up to the share; ×1.4 for the largest	in part on loop: SymJIT plane applications (7db32901), lane-chunked tree closures (88657260); next: the native tree program (A5, native-dag)
Lanes by batch in binary64 (plan_export.lanes_for: 16 lanes also for one point)	$B = 1$: one 16-lane chunk per call ($B = 1$ is reported, not gated: user 2026-10-28)	measured: — at $B = 1$ (16 lanes on loop ÷ 1 lane of 39824f9b, different builds)	executor policy (WP3.1 or wave 4)
Finite-only result path (sum the virtual natively, no per-family lane export, Python assembly only for checks and observables)	glue 3 %–30 % of the native-input time; skipping the pole-only counterterm interactions natively (generation-scale-2's I5 record) saved 2–5 % of the total at $u\bar{u} \to gggg$ and $gg \to gggg$	up to the share (largest at $2 \to 3$ with quarks)	in part on loop (generation-scale-2); next plan_export, executor (wave 4)
Lane-major closures (no scatter, no second copy into the reducer's batch)	scatter 1 %–19 %	up to the share	executor, reducer adapter (wave 4)
AVX2 / AVX-512 kernels (runtime selection)	numerator 39 %–69 %	×1.0–1.4 on the dressing (gate G0)	wave 4 (WP4.6)
NinjaRS and masters: parallel reduction, transposed reduction, stage-2 sharing, a cheaper master cache key	NinjaRS + masters 7 %–29 %; masters 11 %–46 % of the reduction, oneloopmaster alone 1 %–12 %: the cache bookkeeping costs more than the integrals	one thread: up to the share; threads: DESIGN §5.6	NinjaRS requests (DESIGN §11.5)

Below 5 ms per point. $u\bar{u} \to d\bar{d}g$, $u\bar{u} \to u\bar{u}g$, $u\bar{u} \to e^-e^+g$ and $u\bar{u} \to e^-e^+gg$ take 2.20, 2.60, 0.691 and 4.93 ms per point at $B = 128$; by the user's decision of 2026-10-24 a process below 5 ms has no blocking gate and its slowdown against the providers need not be pursued: optimisation effort goes to higher multiplicities. Their ratios stay in the tables ($u\bar{u} \to d\bar{d}g$: 2.20 ms against a limit of 1.97 ms, OpenLoops 1.52 ms × 1.30, 1.12 of it; measured fail, band 1.04–1.19). The question whether G1 should pass the points as exact rationals (the protocol) or as the binary64 text the providers read is moot under this rule, and the protocol stays. For information only (not G1 configurations, no verdict formed from them): on e18a024b the public Runtime.evaluate of the loaded artifact (poles off, core 139) takes 11.19 ms per point at $B = 1$ and 1.91 ms at $B = 128$ for $u\bar{u} \to d\bar{d}g$, against the campaign's 2.20 ms of the harness's executor at $B = 128$ (core 174) and 1.88 ms in its smoke run, on other cores and at other times (public_path_check in data/g1.json, the campaign's STATE.md): the 1.12 of the limit is host noise more likely than a change (1.83 ms at m1028a); at G-int (4e9b5a9e, committed protocol) a paired run (one core, ten alternating rounds) gave 1.68 ms on the binary64 text against 2.29 ms on the exact points (×0.73; $u\bar{u} \to u\bar{u}g$: 2.94 against 3.39 ms, ×0.87).

3.2 Leading colour

Table 8: **Leading colour** (LC), ms per point, same protocol as Table 4; in each cell the full-colour (FC) time of the same configuration and batch, and LC/FC. No provider ratio: the providers time full colour. LC uses the committed QCD colour continuation (`examples/loop/source_colour`); the V+jets rows are timed at LC with native inputs only. This campaign timed LC at $B = 128$; ° grey: the earlier run of `39824f9b` at $B = 1$ with its FC time (an earlier engine under the committed protocol; history).

Process	Python inputs[hist]				Native inputs, `loop`			
	$B = 1$		largest batch		$B = 1$		largest batch	
$gg \to gg$	*not timed*		*not timed*		*not timed*		**0.675**	FC 1.47 · 0.46
$u\bar{u} \to d\bar{d}g$	**40.5**	FC 40.6 · 1.00	**37.4**	FC 32.4 · 1.15	*5.25°*	FC 10.9 · 0.48	**1.61**	FC 2.20 · 0.73
$u\bar{u} \to u\bar{u}g$	*not timed*		*not timed*		*9.00°*	FC 19.1 · 0.47	**1.51**	FC 2.60 · 0.58
$u\bar{u} \to e^-e^+g$	*not timed*		*not timed*		*not timed*		**0.652**	FC 0.691 · 0.94
$u\bar{u} \to ggg$	**184**	FC 215 · 0.86	**171**	FC 179 · 0.96	*22.8°*	FC 50.6 · 0.45	**2.69**	FC 5.26 · 0.51
$gg \to ggg$	**2 350**	FC 2 760 · 0.85	**2 300** ($B = 32$)	FC 2 550 · 0.90	*91.1°*	FC 211 · 0.43	**8.05**	FC 19.5 · 0.41
$u\bar{u} \to e^-e^+gg$	*not timed*		*not timed*		*not timed*		**1.73**	FC 4.93 · 0.35
$gg \to gggg$	*not timed*		*not timed*		*2 070°*	FC 4 100 · 0.50	**176**	FC 384 · 0.46
$gg \to t\bar{t}gg$	*not timed*		*not timed*		*418°*	FC 840 · 0.50	**31.6**	FC 92.2 · 0.34
$u\bar{u} \to gggg$	*not timed*		*not timed*		*380°*	FC 860 · 0.44	**29.0**	FC 73.4 · 0.40
$u\bar{u} \to e^-e^+d\bar{d}g$	*not timed*		*not timed*		*not timed*		**6.41**	FC 16.8 · 0.38
$u\bar{u} \to e^-e^+ggg$	*not timed*		*not timed*		*not timed*		**12.4**	FC 58.1 · 0.21
$gg \to t\bar{t}e^-e^+g$	*not timed*		*not timed*		*not timed*		**11.6**	FC 63.1 · 0.18
$u\bar{u} \to ggggg$	*not timed*		*not timed*		*not timed*		**555**	FC 1 750 · 0.32
$gg \to ggggg$	*not timed*		*not timed*		*not timed*		**3 800**	FC 7 040 · 0.54

LC is cheaper in the loop stage (fewer families and skeletons) and, since lc-perf (`88657260`: LC-aware pruning of the closure blocks, native tree closures in lanes of points), also end to end: with native inputs at $B = 128$ LC/FC is 0.18–0.94 ($gg \to t\bar{t}e^-e^+g$ gains most, $u\bar{u} \to e^-e^+g$ least), so LC is faster than full colour on every row; at G-int it was 0.55–1.28, slower than full colour for $u\bar{u} \to ggg$, $gg \to gggg$, $gg \to t\bar{t}gg$ and $u\bar{u} \to gggg$ (committed protocol; Section 3). The native trees, with the Born and counterterm inputs, take a larger share at LC ($u\bar{u} \to ggg$: 11 % at LC, 7 % at full colour; $gg \to gggg$: 27 % at LC, 26 % at full colour), the lever of compiled trees (Table 7). The $2 \to 5$ rows follow: LC/FC 0.18–0.54 at $B = 128$ ($gg \to t\bar{t}e^-e^+g$ 0.18, $u\bar{u} \to e^-e^+ggg$ 0.21, $u\bar{u} \to ggggg$ 0.32, $u\bar{u} \to e^-e^+d\bar{d}g$ 0.38, $gg \to ggggg$ 0.54). At seven gluons the stored LC plan of $gg \to ggggg$ keeps all 4 707 full-colour loop families, where the lower gluon multiplicities halve them (154 ⟶ 77 and 1 100 ⟶ 550 at five and six gluons, with the top). This is not a defect (DESIGN §7.9): a family is keyed by its propagators' momenta and masses, so a closed massless-quark loop shares its family with the gluon and ghost rings, and the prune drops a family only when none of its skeletons is live; with the top the families halve, under this generation's `particle_veto = [t]` they cannot, while the tries shrink alike. The LC plan generates in 2.9 min against 2.8 min at full colour (`LoopSlice.generate` alone; on `1d27426b` 40.6 against 37.9 min).

Identical-flavour quark lines: since identical-flavour-lc (`5cb5ea83`, milestone m1028a) the leading and next-to-leading colour of processes with identical-flavour quark lines (from $u\bar{u} \to u\bar{u}g$ up, any number of lines) are exact formal-N_c projections: the loop-request blocks and the counterterm program split by U(1)-subtraction N_c grade in the Python and Rust lanes, with order-free block grades (second genericity audit, G2-M1; validated against the legacy engine's LC and NLC, now frozen captures, and since MQ-NC by rule 7 against N_c scans and independent oracles, Section 2). G1′s LC of $u\bar{u} \to u\bar{u}g$ is a validated value (1.51 ms per point at $B = 128$).

3.3 Throughput, model recalibration and schedule M

Table 9: **Kernel throughput and the design recipe** (native inputs, full colour, largest batch). Work: the executor's modelled complex multiply-adds of the chosen kernel forms per point (one lane); throughput: work × lanes over the dressing time (fused-factor builds excluded). Native loop: seeds, factors, dressing, scatter and reduction; outside: everything else of the call (trees, glue). Recipe: DESIGN §1.3, finite only, where the design has a row; measured ÷ recipe for the total and for the NinjaRS stage (masters included) against its model R.

Process	Work, MCMAC/pt	Dressing, ms	G CMAC/s	Native loop, ms	Outside, ms	Total, ms	Total ÷ recipe	NinjaRS ÷ R
$gg \to gg$	0.932	0.678	1.37	1.05	0.423	**1.47**	1.0–2.9	0.97–1.10
$u\bar{u} \to d\bar{d}g$	0.556	0.499	1.12	1.65	0.546	**2.20**	no row	—
$u\bar{u} \to u\bar{u}g$	1.09	0.853	1.28	2.02	0.580	**2.60**	no row	—
$u\bar{u} \to e^-e^+g$	0.168	0.130	1.29	0.441	0.250	**0.691**	no row	—
$u\bar{u} \to ggg$	3.18	2.46	1.29	4.21	1.05	**5.26**	no row	—
$gg \to ggg$	14.3	10.6	1.35	14.5	4.95	**19.5**	no row	—
$u\bar{u} \to e^-e^+gg$	2.89	2.23	1.30	4.16	0.766	**4.93**	no row	—
$gg \to gggg$	268	209	1.28	270	114	**384**	0.9–2.8	1.01–1.19
$gg \to t\bar{t}gg$	56.0	42.5	1.32	73.5	18.7	**92.2**	0.6–1.7	1.01–1.42
$u\bar{u} \to gggg$	56.7	42.6	1.33	61.0	12.4	**73.4**	no row	—
$u\bar{u} \to e^-e^+d\bar{d}g$	8.80	7.58	1.16	14.3	2.51	**16.8**	1.0–3.2	1.03–1.85

Process	Work, MCMAC/pt	Dressing, ms	G CMAC/s	Native loop, ms	Outside, ms	Total, ms	Total ÷ recipe	NinjaRS ÷ R
$u\bar{u} \to e^-e^+ggg$	47.8	37.7	1.27	53.2	4.84	**58.1**	0.9–2.4	0.85–1.18
$gg \to t\bar{t}e^-e^+g$	49.3	38.3	1.29	55.6	7.50	**63.1**	0.8–2.3	1.07–1.48
$u\bar{u} \to ggggg$	1 170	948	1.23	1 460	283	**1 750**	0.7–2.1	1.06–1.44
$gg \to ggggg$	4 060	3 460	1.18	4 110	2 920	**7 040**	0.9–2.6	0.48–0.57

Recalibration. The kernels run 1.12–1.37 G complex MAC/s per core on the real plans, the rate gate G0 measured on its synthetic programs (1.29–1.37); the dressing term of the recipe holds. The NinjaRS stage, masters included, lands on the design's model R (measured ÷ R 0.48–1.85, Table 9). What the recipe misses is the **outer factor**: it assumed trees, seeds and counterterms add 5–10 % (DESIGN §1.3), but with the current native inputs the total is 1.15–2.2 times dressing plus reduction at full colour (up to 3 at LC). Table 10 re-estimates every design row with these measured terms.

Table 10: **Design estimates recalibrated by G1** (`tools/loop_v2_costmodel.py --g1`), ms per point, finite only, one thread. Recipe: DESIGN Table 1a's finite-only band (outer factor 1.05–1.10). G1: the recipe's dressing term, the NinjaRS model R × 0.483–1.849 (the band this campaign measures, Table 9) and the measured outer factor of the row's class representative at $B = 128$ with native inputs (same colour where measured). Measured: G1's own time where the row was timed. Gate: G1's binding limit at the row's largest batch where G1 timed the providers (1.3× the fastest production provider, or MadLoop's time where it binds), else 1.3× the best production provider of DESIGN Table 1b (protocol of 20 October), with the headroom gate ÷ upper estimate. The outer factor shrinks with compiled trees (Table 7); until then these replace the Table 1a targets.

Design row	Col.	Recipe	Outer	Class	G1 estimate	Measured	Gate	Headroom
$gg \to gg$	FC	0.502–1.52	1.52	$gg \to gg$	**0.617–2.30**	1.47	—	—
$gg \to gg$	LC	0.204–0.542	2.07	$gg \to gg$	**0.329–1.16**	0.675	—	—
$gg \to gggg$	FC	139–409	1.44	$gg \to gggg$	**159–600**	384	3 690	6.15
$gg \to gggg$	LC	55.7–146	1.59	$gg \to gggg$	**66.5–246**	176	—	—
$gg \to ggggg$	FC	2 710–7 920	1.74	$gg \to ggggg$	**3 730–14 100**	7 040	—	—
$gg \to ggggg$	LC	1 070–2 830	1.37	$gg \to ggggg$	**1 100–4 110**	3 800	—	—
$u\bar{u} \to ggggg$	FC	827–2 630	1.37	$u\bar{u} \to ggggg$	**929–3 620**	1 750	569 000	157.10
$u\bar{u} \to ggggg$	LC	206–526	1.39	$u\bar{u} \to ggggg$	**206–810**	555	—	—
$u\bar{u} \to e^-e^+ggg$	FC	24.5–68.1	1.15	$u\bar{u} \to e^-e^+ggg$	**21.3–83.5**	58.1	326	3.90
$u\bar{u} \to e^-e^+ggg$	LC	3.34–8.31	1.56	$u\bar{u} \to e^-e^+ggg$	**3.78–14.5**	12.4	—	—
$u\bar{u} \to e^-e^+d\bar{d}g$	FC	5.34–16.5	1.37	$u\bar{u} \to e^-e^+d\bar{d}g$	**5.65–24.3**	16.8	45.9	1.89
$u\bar{u} \to e^-e^+d\bar{d}g$	LC	1.19–3.14	1.81	$u\bar{u} \to e^-e^+d\bar{d}g$	**1.57–6.55**	6.41	—	—
$gg \to t\bar{t}gg$	FC	54.8–156	1.30	$gg \to t\bar{t}gg$	**55.0–215**	92.2	613	2.86
$gg \to t\bar{t}gg$	LC	14.6–40.1	1.53	$gg \to t\bar{t}gg$	**17.2–65.2**	31.6	—	—
$gg \to t\bar{t}e^-e^+g$	FC	27.4–76.1	1.16	$gg \to t\bar{t}e^-e^+g$	**24.2–94.1**	63.1	525	5.58
$gg \to t\bar{t}e^-e^+g$	LC	4.06–11.0	1.58	$gg \to t\bar{t}e^-e^+g$	**4.91–18.6**	11.6	—	—

Table 11: **Schedule M against B** (native inputs, one thread), pure-gluon classes, the only classes where M merges seed entries (DESIGN §5.6). The schedule-dependent stages are seeds, factors, dressing, scatter and the NinjaRS reduction ("the reduction included", §1.6 item 5); the trees, Born and counterterm inputs and the glue are the same work in both schedules. M: the development campaign's runs (`39824f9b`, not repeated); B: the configuration on `loop` (`e18a024b`), a later engine (native tree closures and plane applications, frozen setup heap) on the same processes, so the ratios pair runs of different builds and campaigns (the gate tool pairs them; history for M): end to end at $B = 128$ they set the M time with the older, slower trees against this engine's B time and do not measure M; LC was re-timed at $B = 128$ only. M is adopted where M/B ≤ 0.85 on the schedule-dependent stages at every common batch size; ± is the timing uncertainty of the ratio.

Process	Col.	Stages M/B, $B = 1$	Stages M/B, $B = 128$	End to end, $B = 1$	End to end, $B = 128$	Adopt M
$gg \to ggg$	FC	—	**1.02** ± 0.03	—	2.60	no
$gg \to ggg$	LC	—	**0.94** ± 0.03	—	5.79	no
$gg \to gggg$	FC	—	**0.87** ± 0.06	—	2.36	no
$gg \to gggg$	LC	—	**0.70** ± 0.15	—	5.79	**yes**

B against M (decision, provisional until the trees are compiled). On the schedule-dependent stages M passes the 15 % rule for $gg \to gggg$ at LC (0.70; DESIGN §5.6 measured 0.75–0.80 on the dressing side), and not for $gg \to ggg$ at FC (1.02), $gg \to ggg$ at LC (0.94) and $gg \to gggg$ at FC (0.87). The same rows qualify as in the development campaign's paired runs, on which the decision was made (`data/g1.json` of `c3a7be09`). End to end the gain is diluted by the tree inputs, which both schedules share and which compiled trees will shrink (Table 7). With a quark line M's entries do not merge at all (M/B work 1.00, DESIGN §5.6; equal by test). **Decision (G1):** schedule M for pure-gluon classes at leading colour from $n = 6$; schedule B for every other class at $n \leq 6$, including pure-gluon full colour; M stays the planned default from $n \geq 8$. The seven-leg comparison of M with B is open (the schedule-M run of `measure-g1-7leg` stopped; G1 and Table 1 time schedule B).

Table 12: **Values at the timed points**: pyAmpliCol's binary64 finite part (native inputs on `loop`) against each provider's production-mode value on the common stream points (the providers read the binary64 decimal text of the exact points), largest deviation, metric $|a-b| \,/\, \max(|a|, |b|, |B|\ \alpha_s/2\pi)$; Recola's finite part converted with the universal double pole (it returns no poles in production). Worst

point: the point of the row's largest deviation, evaluated again with one rotation and the reduction-direction check in binary64 and in double-double (DD): E is the binary64 estimate, DD the DD value's deviation from the closest provider (named). Campaign: the first three points against the committed references (`tests/fixtures/loop/required`; $gg \to ggggg$: NJet's ten points), fixture metric and tolerance, asserted in binary64 where its estimate $E \le 10^{-9}$ and in DD elsewhere (user 2026-10-29; earlier records: binary64 throughout), native inputs (`loop`) and Python inputs (history); ‡ points above the tolerance that the user accepted (below the table).

Process	Pts	vs MadLoop	vs Recola	vs OpenLoops	Worst point: E	DD: closest provider	Campaign, native	Campaign, Python[nist]
$gg \to gg$	16	6.7×10^{-13}	3.1×10^{-13}	3.0×10^{-13}	pt 4: 5.1×10^{-12}	9.4×10^{-14} OpenLoops	—	—
$u\bar{u} \to d\bar{d}g$	16	6.7×10^{-13}	3.3×10^{-10}	6.7×10^{-13}	pt 1: 7.8×10^{-12}	3.2×10^{-13} MadLoop	3.6×10^{-10}	3.6×10^{-10}
$u\bar{u} \to u\bar{u}g$	16	8.4×10^{-12}	7.2×10^{-11}	6.1×10^{-12}	pt 1: 7.9×10^{-12}	2.4×10^{-13} MadLoop	8.6×10^{-11}	8.6×10^{-11}
$u\bar{u} \to e^-e^+g$	16	1.3×10^{-13}	8.6×10^{-14}	no library	pt 0: 2.4×10^{-12}	7.5×10^{-15} Recola	—	—
$u\bar{u} \to ggg$	16	1.4×10^{-12}	1.5×10^{-12}	8.4×10^{-11}	pt 2: 1.4×10^{-12}	1.3×10^{-14} MadLoop	—	—
$gg \to ggg$	16	1.0×10^{-6}	1.0×10^{-6}	1.0×10^{-6}	pt 10: 2.0×10^{-5}	6.0×10^{-12} Recola	—	—
$u\bar{u} \to e^-e^+gg$	16	9.4×10^{-11}	4.7×10^{-11}	no library	pt 8: 3.4×10^{-9}	4.3×10^{-13} Recola	—	—
$gg \to gggg$	8	2.2×10^{-7}	2.8×10^{-8}	2.8×10^{-8}	pt 1: 2.1×10^{-6}	4.4×10^{-12} Recola	7.1×10^{-11}	8.7×10^{-8} **above 2.0×10^{-8}**
$gg \to t\bar{t}gg$	16	1.4×10^{-8}	7.9×10^{-9}	7.9×10^{-9}	pt 13: 1.0×10^{-7}	5.5×10^{-15} OpenLoops	1.8×10^{-11}	1.1×10^{-11}
$u\bar{u} \to gggg$	16	—	9.3×10^{-12}	1.6×10^{-10}	pt 9: 5.6×10^{-10}	1.9×10^{-13} OpenLoops	—	—
$u\bar{u} \to e^-e^+d\bar{d}g$	16	4.3×10^{-11}	5.7×10^{-11}	5.7×10^{-11}	pt 15: 1.4×10^{-9}	5.5×10^{-14} OpenLoops	9.0×10^{-13}	—
$u\bar{u} \to e^-e^+ggg$	16	8.9×10^{-11}	1.1×10^{-10}	1.1×10^{-10}	pt 8: 1.6×10^{-9}	7.7×10^{-14} OpenLoops	4.1×10^{-13}	—
$gg \to t\bar{t}e^-e^+g$	16	8.7×10^{-9}	4.4×10^{-9}	5.6×10^{-9}	pt 2: 5.7×10^{-8}	5.8×10^{-13} OpenLoops	1.0×10^{-8}	—
$u\bar{u} \to ggggg$	3	1.9×10^{-12}	2.9×10^{-10}	no library	pt 0: 3.2×10^{-11}	1.4×10^{-12} MadLoop	2.9×10^{-10}	—
$gg \to ggggg$	0	not generated	not generated	no library	—	—	7.1×10^{-9}	—

Reading the values. The Born agrees with every provider to 1.7×10^{-13} or better. For the finite part, the double-double (DD) value at each worst point (Table 12) separates the sources: it agrees with the closest provider to 6.0×10^{-12} or better everywhere. Where pyAmpliCol's binary64 estimate E exceeds the deviation — $gg \to ggg$, $gg \to gggg$ and $gg \to t\bar{t}gg$ (above 10^{-9}) and $u\bar{u} \to gggg$ (1.9×10^{-13} in DD against 1.6×10^{-10} in binary64 at point 9) — the deviation is pyAmpliCol's own binary64 loss, which E flags. The other deviations above 10^{-11} stay in DD and are the providers' own accuracy: Recola's at $u\bar{u} \to d\bar{d}g$ (3.3×10^{-10}), $u\bar{u} \to u\bar{u}g$ and $u\bar{u} \to e^-e^+gg$ (COLLIER's default target 10^{-8}, HANDOFF §7), OpenLoops' at $u\bar{u} \to ggg$ (8.4×10^{-11}) and MadLoop's at $u\bar{u} \to e^-e^+gg$ (1.1×10^{-10}); MadLoop's production values also stay off the DD values at the ill-conditioned gluon and top points (up to 2.4×10^{-7}). At the campaign points the check asserts binary64 where its estimate $E \le 10^{-9}$ and double-double elsewhere (user E1 ruling 2026-10-29 ~03:45, package g1-accuracy): $gg \to gggg$ point 1 (E = 2.1×10^{-6}) is asserted in DD, the check passing at 7.1×10^{-11} (tolerance 2×10^{-8} binary64 there 1.5×10^{-8}, recorded); its DD value agrees with Recola to 4.4×10^{-12}. ‡ $gg \to ggggg$ against NJet: points 4 and 8 exceed the tolerance 2×10^{-8} in binary64 (4.4×10^{-8} and 2.1×10^{-8} from NJet's double-double values, at most 7.1×10^{-9} over all its references), **accepted (user 2026-10-25): O(1e-8) binary64, within the estimate, DD = NJet**: estimates E 1.7×10^{-6} and 3.3×10^{-7}, and pyAmpliCol's double-double values equal NJet's; the check keeps its tolerance and, asserting DD at these points ($E > 10^{-9}$, E1 ruling), passes.

4 Required processes: references and development validation

The plan requires the $2 \to 5$ processes of Table 1 (plan §5.7), compared with MadLoop and Recola, and with OpenLoops where a public library exists. A reference provider is run only if its generation completes in about 45 minutes, or in a time comparable to pyAmpliCol's own generation of the process (HANDOFF §0.1, addendum F). Otherwise pyAmpliCol is validated by Ward and gauge identities, Catani poles, LC/NLC/FC and DD/AP consistency, stability checks, lower multiplicities and existing references.

Table 13: **Provider generation on this host and the 45-minute rule** (HANDOFF §0.1): the rows of Table 1 where a provider generation takes more than 45 minutes or did not finish. MadLoop: graph generation + source export + compilation; Recola: process generation; OpenLoops: prebuilt public libraries (no generation time). Records: `data/g1.json` (G1 rows) and `data/measured.json` (`provider_generation`, the reference runs of the $2 \to 5$ rows), through `data/summary.json`. Every other row of Table 1 generates within 45 minutes with every provider (longest: MadLoop 10.9 min for $gg \to t\bar{t}e^-e^+g$ (EW)).

Process	MadLoop	Recola	OpenLoops	45-minute rule
$gg \to gggg$	2.4 h (285 + 8 354 + 37 s)	35.4 min	prebuilt ppjjjj	**MadLoop over 45 min; results exist and are used**
$u\bar{u} \to ggggg$	12.2 h (5 793 + 38 040 + 122 s)	2.8 h	no public library	**MadLoop, Recola over 45 min; results exist and are used**
$gg \to ggggg$	diagrams 17.4 h; unfinished (last state: exporting)	unfinished (last state: running, 2026-10-20)	no public library	**stopped 22 Oct; not to be restarted (HANDOFF section 0.1)**

Table 14: **Native validation of the required set** DEV (`work/v2-generation-scale` before its integration into `loop` at `4e9b5a9e`, DESIGN WP3.1: ninth increment (`6f5a64cc`; threads not recorded) for $u\bar{u} \to e^-e^+d\bar{d}g$, $gg \to gggg$, $gg \to ggggg$; eighth increment (`0ea83af2`; threads not recorded) for $u\bar{u} \to e^-e^+ggg$, $gg \to t\bar{t}e^-e^+g$ (QCD), $gg \to t\bar{t}gg$; seventh increment (`fef92189`; threads not recorded) for $u\bar{u} \to ggggg$): generation (`LoopSlice.generate`, then the plan export; peaks of each stage; shared host under load), then binary64 virtuals with native per-point inputs and the production counterterm route at the fixture points, against the references committed on that branch (`tests/fixtures/loop/required`; fixture tolerance 2×10^{-8}). Largest deviation over the references and points, metric of Table 1; the per-reference values are in `data/measured.json`. For $gg \to ggggg$ the references are NJet (double-double and two binary64 runs) and the Catani operator for the poles, top-free with $N_f = 5$; its estimate column is the rotation plus reduction-direction check per point.

Process	References	Born	Finite	$1/\varepsilon$	$1/\varepsilon^2$	Estimate E	Generation + export
$u\bar{u} \to e^-e^+d\bar{d}g$	ML, Rc, OL · 3 pts	1.9×10^{-14}	8.7×10^{-13}	2.3×10^{-13}	1.0×10^{-13}	1.2×10^{-12}, 2.8×10^{-12}, 1.1×10^{-12}	37.7 s 0.4 GiB
$u\bar{u} \to e^-e^+ggg$	ML, Rc, OL · 3 pts	2.3×10^{-14}	8.5×10^{-13}	4.6×10^{-13}	2.0×10^{-13}	—	36.7 s 0.5 GiB
$gg \to t\bar{t}e^-e^+g$ (QCD)	ML, Rc, OL · 3 pts	1.4×10^{-14}	5.7×10^{-9}	3.7×10^{-12}	5.3×10^{-12}	—	43.7 s 0.6 GiB
$gg \to t\bar{t}gg$	ML, Rc, OL · 3 pts	2.1×10^{-14}	1.1×10^{-11}	1.0×10^{-10}	1.6×10^{-11}	—	38.5 s 0.7 GiB
$gg \to gggg$	Rc, OL · 2 pts	4.7×10^{-14}	4.5×10^{-11}	1.2×10^{-11}	5.2×10^{-12}	1.5×10^{-10}, 1.4×10^{-10}	154 s 2.2 GiB
$u\bar{u} \to ggggg$	ML, Rc · 3 pts	1.7×10^{-13}	2.9×10^{-10}	6.1×10^{-10}	4.4×10^{-11}	—	517 s 6.1 GiB
$gg \to ggggg$	NJet DD, NJet R1, NJet R2, Catani · 1 pt	8.1×10^{-13}	3.8×10^{-9}	1.5×10^{-9}	1.1×10^{-12}	2.0×10^{-8}	51.5 min 28.7 GiB

The $gg \to t\bar{t}e^-e^+g$ (QCD) finite part deviates by 5.7×10^{-9} at one of its three points and by at most 3.7×10^{-12} at the others: the point's known binary64 conditioning (earlier record, DESIGN WP3.1 fourth increment: $\le 3.6\times10^{-12}$ at two points; third point 5.7×10^{-9} in binary64 (estimate 1.7×10^{-9}), MadLoop to 1.9×10^{-14} in DoubleFloat). The electroweak target waits for M-SM (WP3.2, paused: EW guards, the QED Catani operator, EW fermion lines, Feynman-gauge trees, loops and counterterms (EWI-4a, EWI-4b), on-shell external photons at $\alpha(0)$ (EWI-6b), the UFO's EW counterterms lowered at scale (EWI-0, EWI-3, EWI-3R), Levi-Civita cells (EWI-7b) and lepton lines through the loop (EWI-5) are on `loop`; EW native inputs (EWI-N), the EW ladder and the target (EWI-7, EWI-9) are not; the QCD classes are M-QCD's, Section 2). The $u\bar{u} \to ggggg$ row comes from DESIGN.md WP3.1 seventh increment (fef92189), the newest record of that process: the later re-runs on the merged heads (eighth and ninth increments) do not include it. It uses the same production counterterm route; its check is in Table 16.

5 Seven gluons: NJet and pyAmpliCol

5.1 NJet: Table 5 of arXiv:1209.0100, re-measured on this host

NJet is the independent reference for $gg \to ggggg$, where MadLoop and Recola cannot generate under the 45-minute rule and OpenLoops has no library. It is massless five-flavour QCD, and it is not one of the gate's matched providers.

Table 15: **NJet, s per point**, double precision, two evaluations each (NJet's scaling test). Paper: Xeon E3-1240 at 3.3 GHz, NJet 1.0. This host: one pinned core, idle at launch, of the shared host; 10 frozen points per channel, mean (spread ≤ 2 %); the multiplier is time ÷ the paper's. De-symmetrised sums equal the standard sums only after phase-space integration; they are timing figures, not point values.

Channel	Table 5	NJet 3.1.2		NJet 2.1.1		Table 5	NJet 3.1.2		NJet 2.1.1	
	standard full-colour sum					de-symmetrised sum				
$gg \to gg$	0.03	**0.00295**	×0.1	**0.00291**	×0.1	0.03	*no class*		*no class*	
$gg \to ggg$	0.22	*defective channel*		**0.141**	×0.6	0.07	*defective channel*		**0.0452**	×0.6
$gg \to gggg$	6.19	**4.59**	×0.7	**4.58**	×0.7	0.57	**0.39**	×0.7	**0.38**	×0.7
$gg \to ggggg$	171.31	**142**	×0.8	**246.7**	×1.4	3.07	**2.41**	×0.8	**4.19**	×1.4

At seven gluons one NJet 3.1.2 evaluation without the scaling test takes 71.2 s (standard) and 1.22 s (de-symmetrised; means over 10 points), at 69 MB; double-double costs 1 990 s per point. NJet 3.1.2's numerical five-gluon channel is 9/2 too large (NJet 2.1.1 is used there), and at the unrotated seven-gluon points NJet keeps only 4–6 digits in double precision (NaN at 3 of 10), so the reference values are taken at exactly rotated points.

5.2 pyAmpliCol at seven gluons DEV

On the WP3.1 branch before its integration (G-int; Table 14) the full $gg \to ggggg$ virtual, with the counterterms of the recursion and native per-point inputs, top-free with $N_f = 5$, agrees with NJet's double-double values to 3.8×10^{-9} in the finite part (per point: 3.8×10^{-9} over all references, point 0, ninth increment) and to 1.1×10^{-12} in the double pole; each point's rotation plus reduction-direction estimate bounds its difference. Generation (ninth increment) took 44.2 min at 14.7 GiB and the plan export 7.3 min at 28.7 GiB (peak RSS of each stage; 160 MB of plan JSON; the whole validation run, point evaluations included, peaked at 38.6 GiB), inside addendum H's budget (hours, below 150 GB). DESIGN WP3.1 attributes the cost: 325 s before the counterterm recursion, 1,122 s with its counterterm program, 2,766 s once poles-uvir's always-generated companion merged (eighth increment), and 2,652 s after the ct-simplify merge (ninth increment, this record), 1,843 s of it the companion's seed table, which built generic seeds. Since then native-seeds' in-crate colour reductions and generation-scale-2's lean companion (sharing the slice's certificates) brought the companion's seeds to 4.7 s (DESIGN WP3.1), and G1's stored process on `loop` (`e18a024b`, full colour) generates in 2.8 min plus 3.6 min of export at a 9.5 GiB peak (one pinned core, cold caches; Table 4). A point with its rotation and reduction-direction checks took 82.3 s, a single-point development evaluation (threads not recorded): not a batched one-thread time and not comparable with NJet's 71.2 s per evaluation on one pinned core. The batched one-thread timing is G1's (Table 1, Section 3), and so is the check of the timed configuration against NJet's ten points (Table 16).

5.3 Validation without a matched provider

Table 16: **Checks of the seven-leg QCD results.** First and third rows: G1's checks of the timed configurations (engine `e18a024b` on `loop`, native per-point inputs, full colour; binary64 finite part and Born, and the double pole of G1's physical rows against the universal one); second row DEV on `work/v2-generation-scale`; Ward row WP2.5 on `loop`; deviation metric of Table 1. NJet values: fixture `gg_ggggg_njet.json` (NJet 3.1.2 double-double at exactly rotated points, scaling-test estimates ≤ 1.3×10^{-24}; points 0–2 are G1's stream points 0–2 at NJet's scale).

Check	Result	Status
$gg \to ggggg$ full virtual against NJet (top-free, $N_f = 5$), 10 points (G1)	finite ≤ 4.4×10^{-8} against NJet's double-double values (median 3.2×10^{-9}; NJet's binary64 values ≤ 7.1×10^{-9}), Born ≤ 9.3×10^{-13}; every point within its estimate E; asserted in DD at points 0, 1, 4, 5, 6, 7, 8, 9 ($E > 10^{-9}$, E1 ruling); above the tolerance: point 4 4.4×10^{-8} ($E = 1.7\times10^{-6}$) and point 8 2.1×10^{-8} ($E = 3.3\times10^{-7}$), where the double-double values equal NJet's; double pole against the universal one ≤ 5.1×10^{-12} (points 0, 1, 2)	passed (tolerance 2×10^{-8})
Top-free full virtual against NJet at lower multiplicity	$gg \to gg$: ≤ 2.1×10^{-14} at 10 points, full virtual; $gg \to ggg$: ≤ 2.3×10^{-13} at the stable points; the flagged point 3×10^{-14} in DoubleFloat; $gg \to gggg$: 1.0×10^{-11} at point 0; point 1 1.6×10^{-7} in binary64 (estimate 2.7×10^{-6}), 4.7×10^{-15} in DoubleFloat (`test_loop_v2_topfree.py`)	passed
$u\bar{u} \to ggggg$ full virtual against MadLoop and Recola, 3 points (G1, the timed stream points)	finite: MadLoop ≤ 1.9×10^{-12}, Recola ≤ 2.9×10^{-10} (Recola's own: pyAmpliCol's double-double value lies 2.9×10^{-10} from it and 1.4×10^{-12} from MadLoop); Born ≤ 1.7×10^{-13}; double pole against the universal one ≤ 6.8×10^{-13}; the providers agree to 6.2×10^{-10} (reference merge)	passed (tolerance 2×10^{-8})
Ward identity, one gluon's $\varepsilon \to p$, at seven gluons	MQ-WARD (`d3081568`): the generic gauge-Ward check on the native per-point input path, top-free $gg \to ggggg$, legs 1, 4 and 7: virtual ≤ 2.9×10^{-10} (binary64) and 2.8×10^{-26} (DD) of the physical scale, Born ≤ 9.8×10^{-16} / 5.3×10^{-32} (nightly `test_loop_v2_ward.py`; `data/milestone.json` `status_notes`)	**passed (MQ-WARD) at seven gluons (addendum H)**

6 Capabilities at this milestone

Table 17: **Capability matrix** of the v2 engine on `loop` at e18a024b, with branch work marked.

Feature	Status	Boundary
Recursive engine	supported	Trees, families and loop numerators from recursions; no loop diagram is enumerated (FeynKit graphs are test oracles only). Any number of coloured fermion lines, identical or distinct flavours, QCD and EW (fermion-lines, db1ed5cb: the fold of mixed colour-connection parities before the evaluation reuse, Python and Rust lanes; $u\bar{u} \to d\bar{d}s\bar{s}$, $u\bar{u} \to u\bar{u}d\bar{d}$, $u\bar{u} \to u\bar{u}u\bar{u}$ and $u\bar{u} \to u\bar{u}e^+e^-$ equal MadLoop in quadruple precision and Recola). Since ew-leptons (EWI-5, b82d8280) any number of lepton and colour-singlet fermion lines, through the loop too, with one reading-sign rule and one fermion classifier taken from the tree engine (Bhabha against MadLoop; genericity audit 2 G2-M5, G2-M6); since ew-epsilon (6da37352) one root-closure reading sign on every fermionless vertex (G2-M7). Listed, typed refusals replace every silent-wrong-value path of the second genericity audit (fail-closed, fb54246c; visible in the dry run). Counterterms: see below.
Default engine	supported	v2 is the only one-loop engine (MQ-LEGACY-REMOVE, 539dffdd; user 2026-10-28 ~16:10 Q3): the per-diagram legacy engine, its plan v3 artifacts and the precision-rescue options are removed; `loop.engine`, `--loop-engine` and a non-default `[evaluation.stability]` are refused with the removal reason; tests that used the legacy engine as an oracle keep an independent one (frozen captures or providers).
Stock `loop_sm`, public route	supported	The wheel ships MG5 3.7.0′s `loop_sm` and its restrictions with a `[loop.model]` profile (MQ-MODEL, 321efdb4): every QCD class generates through `Generator` / TOML / CLI without a `tools/` helper or hand-written declaration, a multiparticle label one entry per subprocess, bitwise the tools route (Section 2); MadLoop syntax mapped (`[virt=QCD]` → `loop_coupling_orders`, QED=n → `max_coupling_orders`, `/ x` → `particle_veto`), decays, decay chains and $ refused typed; `pyamplicol self-test` evaluates a `loop_sm` virtual against MadLoop.
Full, NLC and LC colour	supported	**Full colour is the default** of loop requests (user 2026-10-28 Q1); LC and NLC are opt-in through the shipped `loop_sm` N_c declaration (MQ-NC, d8d9349f: every counterterm vertex, resolved by structural digest; formal-N_c projections equal to N_c scans and checked against independent oracles, rule 7; an uncovered vertex refused, `colour-declaration-coverage`). **LC helicity and colour-flow selection** (MQ-SELECT, e4964fc3): `helicities=` / `color_flows=` at generation (recursion pruned to the selection) and at evaluation, tree-engine IDs; a complete set of selections sums to the inclusive LC value; NLC and full colour keep their API (selectors refused there). Exact seed tables in N_c, built in Rust since native-seeds (7e3841f3), with native colour cells, seed records and dipole orbits since native-cells (5ebcebea; the Python stages are the oracles); LC/NLC by generated classes and truncated cells. LC is faster than full colour end to end on every G1 row at $B = 128$, the $2 \to 5$ rows included (LC/FC 0.18–0.94, Table 8); the top-free seven-gluon LC plan keeps every full-colour family by construction (Section 3.2). Since identical-flavour-lc (5cb5ea83) LC and NLC of identical-flavour quark lines (from $u\bar{u} \to u\bar{u}g$ up, any number of lines) are exact formal-N_c projections: blocks and counterterm program split by U(1)-subtraction N_c grade in both lanes, order-free grades (G2-M1; the limitation of m1026a is removed, Section 3.2); a block that still mixes grades is refused (`colour-accuracy-nc-exact`, fail-closed).
FFT colour contraction	supported	Certified symmetric-group FFT of the seed contraction in the native executor at full colour and NLC, binary64, double-double and MPFR (MQ-FFT, da6980b0): generic over the plan's certified pairs, policy `auto | direct | fft` (the algorithm run is reported; a forced FFT without a certified pair is refused, `forced-seed-fft-unserved`), public `color.contraction` route; FFT = direct to rounding up to $gg \to ggggg$; the seed stage ×0.46–0.70 of direct at S_5, S_6, slower below, the whole evaluation unchanged within noise (Table 3).
Colour and spin correlations	supported	Directed colour histories at LC/NLC/FC, spin sources, resolved helicities, flavour aliases, Ward sources (WP2.5). Since generation-scale-2 the directed metrics are native and stored with the process (correlated and resolved calls of a stored process regenerate nothing; one seed column per self-adjoint dipole): declaring the NLO dipoles costs +10 % of the six-gluon generation (38.5 against 35.0 s); at seven gluons 784 against 419 s, whose metric tables go to `native-cells`' next run. LC/NLC correlations of identical-flavour quark lines use the graded blocks of the colour row. Correlations keep their internal validation; independent provider oracles (MQ-CORR) are M-FINAL's (user 2026-10-28 Q2); correlations with an LC selection are refused (`selected-correlators`).
Per-order partials	supported	Every squared coupling order at tree level and at one loop, mixed Born orders, per-correction loop orders.
Poles at runtime	supported	`poles = "all" | "none" | "uv" | "ir" | "split"` (WP2.10, on `loop` since 24 October): finite-only mode with the finite part bitwise unchanged and the pole work skipped (natively per counterterm interaction since generation-scale-2's I5 record); `"uv"` and `"ir"` with the user's semantics of 23 October, from a lean stored companion of the certified-scaleless families (bitwise at $gg \to ggg$ and $u\bar{u} \to gggg$, equal to rounding at six and seven gluons) and in the SDK v4 too; the IR poles sum one Catani operator per correction, QED included (ew-poles, 8b52179b; a correction without a prediction gives none); since e6e7543a a mixed QCD×EW sector is predicted only where the requested corrections complete its loops, and joint requests predict it exactly; since MQ-EWJOIN a sector whose QCD loop selection omits IR-divergent loops (gluon exchange between quark lines joined by a W/Z in the loop, `separated-exchange`) is not predicted and checked against MadLoop per split order. Photon-only QED corrections on `loop_qcd_qed_sm` are refused (`veto-unlabelled-counterterms`: the veto reaches counterterms with empty or nominal loop labels; full EW on that model matches Catani; EWI-7). The UV residue comes from an interim local module until the NinjaRS re-pin; a `"uv"` estimate needs a rotation check.
Stability estimates	supported	**Recommended checks** (MQ-STAB, a2aa0d84): `loops.v2.stability.RECOMMENDED_CHECKS`, one generic exact rotation and the rescaled reduction check, the O1 estimate of the campaign and G1 (calibration: binary64 misses 0 % / 0.05–0.10 % of points, DD none; true error ≤ 10 E at every campaign point; R3, DECISIONS 2026-10-29 ~18:25). Rotation checks (any number, 0 = estimate absent), reduction direction, exact rescaling, in any combination; no automatic escalation and no rescue (removed with the legacy engine). Generation's post-build validation re-checks a binary64 miss alone against a 32-digit reference: double-double must agree at the same

Feature	Status	Boundary
		tolerance, binary64 Borns at the tolerance and binary64 virtuals within the larger of the tolerance and their own estimate (one exact rotation and the reduction direction; an estimate above 10^{-9} or absent refuses), else generation refuses; an accepted point is logged (onshell-consistency, 4fc72e39). The rotation and reduction-direction checks flag the fast binary64 lanes' known misses (precision row): use them for such kinematics or model units.
Precision carriers	partial	Binary64, DoubleFloat, arbitrary precision and exact rationals in the native executor; since G-int the native per-point inputs run in binary64, DoubleFloat and arbitrary precision (DESIGN §5.5). Since 7db32901 binary64 runs SymJIT plane applications and needs no Symbolica licence; DoubleFloat and arbitrary precision run under pyAmpliCol's library key. Since precision-ap (a8409830) a master-integral invariant of one external leg is that leg's squared mass at the point's parameters: DoubleFloat and arbitrary-precision virtuals were O(1) wrong at binary64-rounded massive momenta and now agree with a MadLoop quadruple-precision run within 10^{-15} (and DoubleFloat massive spinors along $-z$ are fixed); since generation-scale-2 derived masses (e.g. M_W of loop_sm) are parameter-program outputs, bound exactly (genericity M2). Since onshell-threshold (6e123fdb, G2-M3) binary64 lanes at the fast engine's absolute thresholds (light masses, invariants near a mass shell) go to the generic binary64 engine. **Known limitations of the fast binary64 lanes** (NinjaRS spinor package paused): Ninja's light-like spinors are off by up to $\sim 10^{-5}$ for legs near the $-z$ axis, and models whose units are $\sim 10^{4}$ or more away from GeV can give silent percent-level errors with stability checks off; double-double is exact there and the recommended checks flag them.
Native executor, schedules	supported	Rust, 16 lanes (also for a one-point call, a G1 lever), threads; native tree closures in lanes of points with runtime AVX-512/AVX2 widths (lc-perf); in finite-only mode the counterterm interactions that only feed poles are skipped natively (generation-scale-2); G1 decision on the schedule-dependent stages (Table 11, provisional until compiled trees): M for pure-gluon LC from $n = 6$, B for every other class at $n \leq 6$; F as oracle. The NinjaRS reduction runs serially. Development report: stage times, modelled kernel work, master-integral time.
Plan v4 artifacts and API	partial	Published and loaded through Python, TOML and CLI; the artifact stores the native process (point_inputs = "native-v2"). Since 7db32901 (addendum G) its SymJIT -O2 applications are stored as IR and recompiled at load for the loading CPU (AVX and FMA3 required), no host string is embedded, and binary64 values are bitwise equal after relocation to another machine, user and path; the parameter program's SymJIT bytes differ per compilation (values unaffected; upstream fix filed). Since model-determinism (4e9262b5) the compiled model and the artifact's digest no longer depend on the process history; since native-lowering (79f2a5bd) the plan JSON is written by the Rust plan model (Python writer as oracle). Since generation-scale-2 (11542fd5) no call of a stored process regenerates (every pole selection with the stored companion, correlated, resolved, flavour and catalogue calls), a process the native export cannot store fails its generation with a typed refusal (genericity M1; the plan-only, regenerated-at-load artifacts are gone), and the Rust/C/C++/Fortran SDK v4 evaluates these artifacts with the "uv"/"ir" selections, bitwise the Python API. Since sdk-parity (f6619f34) one parameter rule holds in the Python API, V2Runtime, the Rust SDK and the C/C++/Fortran clients: ASCII decimal values checked before names, unknown, generation-fixed (naming who fixed them) and scale parameters refused with typed errors (G2-M2). The release natives need glibc $\geq$ 2.38, a lower glibc giving the typed NativePlatformError at load (glibc-floor, 5301e24a). Open: external-photon artifacts wait for EW native inputs (EWI-N); the plane check's outcome depends on the parameter order (G2-M4: a listed plane-mismatch refusal, never a wrong value).
Counterterms (model UV, R_2)	supported	Plain tree vertices of the recursion inserted exactly once (WP2.2, a18b7599; addendum J, b4d76cc3), admitted by coupling orders and the public particle_veto; no witness table or loop matching in production. **Stock loop_sm compatibility rule** (MQ-MODEL): its four-G^0 vertex V_1001 names the three-slot constant SSS1 where the four-slot SSSS1 (same exact value) is meant; model preparation moves such references by structure, only on vertices a request reaches, and records each repair (pyamplicol_source_repairs, source_compatibility_diagnostics in the dry run); any other slot mismatch is refused (lorentz-slot-arity). Vertices whose couplings vanish at the restriction leave the loop catalogue (fail-closed-2, loop-vertex-outside-tree-model otherwise). An unlowerable counterterm vertex that no tree of the request reaches is inert and recorded in the artifact; a reachable one is refused. Site closure is the test oracle. Per point they are formed natively (G-int), the Python stages kept as oracle; generation of the counterterm route has output-identical savings (ct-perf, 201bece5: compile memos, a content-addressed lowered-model cache), and since A4 (1b13005a) the native DAG builder is the production route for the loop blocks and the counterterm program (its counterterm emission byte-identical to the Python oracle; 11 listed features keep a recorded per-request Python fallback, any other native failure fails closed), at parity with the site oracle. In EW requests ('t Hooft–Feynman gauge, EWI-4b, 8b19dae8) the Goldstone–vector mixing counterterms are lowered as vertex insertions. The UFO's EW counterterms (ew-ct, 392d3f4a) are lowered through the model's own helper bodies with MadLoop's branch side for exactly real arguments: every Laurent piece of MadLoop's model library equals pyAmpliCol's to 3.7×10^{-15} in binary64 and 6.0×10^{-31} in double-double (against MadLoop's quadruple precision). Since EWI-3R (59d47cb3) the lowering runs at scale: native, parallel kernel-equivalence proofs (Python the oracle), one reference pair per Laurent coefficient, shared Horner schemes; about 26 s cold at the target instead of 259 s, 8.4 % of the G_μ $gg \to t\bar{t}$ (EW) generation (criterion < 20 %, user 2026-10-28); a non-polynomial coupling term or one below ε^{-2} is refused when reachable. Lorentz Epsilon is lowered (EWI-7b, 6da37352; sign = MadLoop to 10^{-12}), and the Goldstone–vector mixing sign, from the tree engine's momenta, matches MadLoop's counterterm selection.
Direct R_2	deferred	A bonus by the user's rule; bounded controls only.
Electroweak, complex mass	deferred	WP3.2. On loop: typed refusals of EW requests with a non-zero width (until the complex-mass scheme); the QED Catani operator (8b52179b); EW fermion lines (1d27426b): Dirac–Weyl EW vertices, W vertices along the fermion flow, conjugate readings, lepton- and colour-singlet-rooted processes, mixed-order Born with W exchange; 't Hooft–Feynman gauge for

Feature	Status	Boundary
		trees (EWI-4a, 6c87e276; Feynman = unitary and Slavnov–Taylor identities to 10^{-13}; two model-compiler defects fixed, also on main: the chirality of a lone projector, 3ae34543, and the Goldstone flag of G^-) and for loops and counterterms (EWI-4b, 8b19dae8: W, Z, Goldstone and ghost loop lines, Goldstone–vector mixing counterterms; QCD-only requests keep the compiled gauge, plans byte-identical; $gg \to t\bar{t}$ (EW) = MadLoop within 10^{-12}); on-shell external photons at $\alpha(0)$ with MG5_aMC's conversion (EWI-6b, e6e7543a); the UFO's EW counterterms (EWI-0, EWI-3) lowered at scale (EWI-3R, 59d47cb3, above); Lorentz Epsilon and one reading sign per vertex (EWI-7b, 6da37352; [U2] anomalous vetoes, which break a doublet, refused by default with a recorded override); lepton and colour-singlet fermion lines through the loop (EWI-5, b82d8280: Bhabha against MadLoop, $gg \to t\bar{t}e^-e^+$ (EW) against three providers, $e^-e^+ \to t\bar{t}h$ against Recola; MadLoop's export omits the R_2 term R2_G0tt_b, shown from its own output and with Recola). Refused (typed): photon-only QED on loop_qcd_qed_sm, charged Goldstone mixing reached only by $gb \to t\bar{t}b$, ghost vertices not closed under conjugation, $u\bar{d} \to t\bar{b}$ (EW), Goldstone mixing at a massive leg (EWI-7) and the complex-mass scheme (EWI-10). Paused until M-SM: EW native inputs (EWI-N), the EW ladder and the target (EWI-7, EWI-9); CMS (EWI-10).
Required set	**partial**	Seven of eight generate and agree with their references in all coefficients with native inputs (Table 14, validated on the WP3.1 branch before its integration), $gg \to ggggg$ against NJet's double-double values (its binary64 points 4 and 8 at $O(10^{-8})$ accepted by the user, within the estimate); the generator is on loop since G-int, and G1 times all seven in batches at full and leading colour (Section 3; the $2 \to 5$ rows since measure-g1-7leg; all re-timed at e18a024b). The seven-gluon Ward check (addendum H): passed (MQ-WARD). The G1 accuracy check asserts binary64 where $E \le 10^{-9}$ and DD elsewhere (user E1 ruling, g1-accuracy); R3 for QCD is recorded (DECISIONS 2026-10-29 ~18:25: two providers per process within their agreement, $gg \to ggggg$ against NJet alone, addendum F). The EW target waits for M-SM (WP3.2). The provider reference trees no longer read the old working root (reference-relocate, 48aeb123; provider values bitwise, and the toolkit refuses provider files outside the configured reference root).
Performance gates	**partial**	G1 on loop (Section 3, native per-point inputs, e18a024b) at $B = 128$: 9 of 9 gated processes pass; 4 below 5 ms per point are not gated (user, 2026-10-24); $gg \to ggggg$ without a matched provider: no gate; the gate is taken at $B = 128$ only (user 2026-10-28): $B = 1$ and the $2 \to 5$ rows' $B = 16$ are reported without a verdict (Table 5). The Python per-point inputs before G-int failed every gate they generated (history). Missing: provider protocol-mode columns.
$gg \to 6g$	**deferred**	No $2 \to 6$ work until the $2 \to 5$ plan is complete (user, 23 October).

7 Architecture of the v2 engine

7.1 The computed quantity

For Born order a, loop order b and squared coupling order $s = a + b$,

$$V_{s,k} = \mathcal{N} \sum_{a+b=s} 2\,\mathrm{Re}\left[B_a^\dagger G_{BL} L_{b,k}\right], \quad k \in \{0,-1,-2\}, \quad V = C_\varepsilon \left(V_0 + \frac{V_{-1}}{\varepsilon} + \frac{V_{-2}}{\varepsilon^2}\right), \quad C_\varepsilon = \frac{(4\pi)^\varepsilon}{\Gamma(1-\varepsilon)},$$

where G_{BL} is the colour metric between Born and loop colour structures and $\mathcal{N}$ holds the initial-state averages and identical-particle factors (MadLoop/BLHA convention; couplings included, nothing divided by the Born). Directed colour and spin correlations replace $2\,\mathrm{Re}(\ldots)$ by the two cross terms $B_A^\dagger G L_B + L_A^\dagger G B_B$.

7.2 Pipeline

For every denominator family f, sector s and observable column o, v2 builds the four-dimensional numerator $N_{f,s,o}(q) = \sum_\alpha c_\alpha q^\alpha$ that NinjaRS reduces: Born-weighted and helicity-summed, produced by three recursions that share all work (Figure 2).

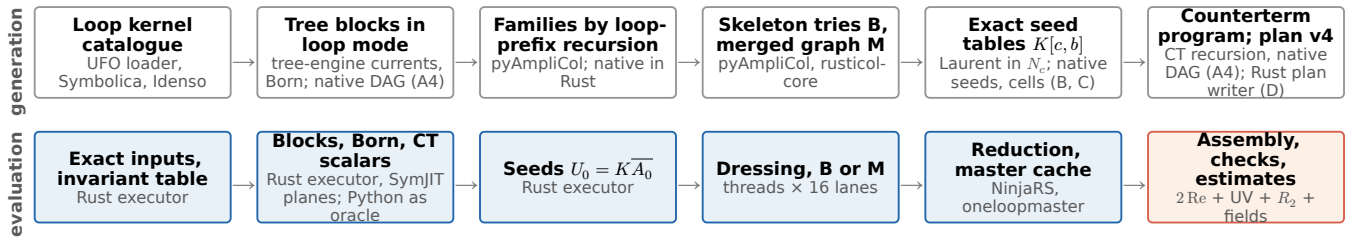


Figure 2: **v2 pipeline.** Upper lane: generation, once per process, into a plan v4 artifact. Lower lane: evaluation per batch of points. Blue: native batched numerics in the Rust executor (the per-point inputs since G-int); orange: Python per batch; white: generation. The owning code or dependency is named under each step.

Trees. The tree engine's colour-ordered Berends–Giele recursion computes, once per point, every off-shell current over every leg subset a loop vertex can absorb (the *blocks*), the Born colour components A_0 and, with WP2.2, every tree with one counterterm vertex. A loop-aware reachability test keeps exactly the currents some family position or root needs.

Blocks with transverse sources use the exact transverse Yang–Mills kernels; loop segments, counterterm currents and Ward sources use the full ones.

Families and skeletons. A Recola-type recursion over loop prefixes (ordered leg sets, propagator species and masses) yields the denominator families, keyed by the canonical cyclic sequence of (leg mask, mass) and equal to Recola's tensor-integral identities. Per family it yields the colour-ordered *skeletons*: rings of blocks, each block summing all trees on its leg subset (Figure 3). Skeletons are never stored as a list; they are the leaves of the schedule tries. Only certified structural zeros are dropped (scaleless rings, external wave-function corrections, VEV tadpoles, colour zeros); nothing is restricted to 1PI.

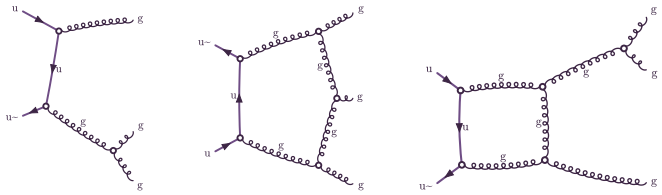


Figure 3: Three graphs of $u\bar{u} \to ggg$ drawn by FeynKit's renderer from its own generation (an oracle, not a production input). In v2 the Born (left) is a closure of blocks; the pentagon (middle) is a ring of five single-leg blocks; the box with a tree attachment (right) is a ring of four blocks, one of them the two-leg current, computed once per point and shared by every skeleton that contains that leg subset.

Loop numerators and schedules. An open-loop recursion dresses skeletons one segment at a time with polynomial-valued objects, $O' = O \cdot Y + \mathrm{raise}\big(O \cdot Z_\mu, q^\mu\big)$, in the NinjaRS monomial layout ($\binom{r+4}{4}$ coefficients at rank r, no rank cap). The seed is the colour-summed Born interference, and each block's helicities are summed as it is attached. One generator feeds three schedules (Table 18).

Table 18: **Three schedules on one generator** (DESIGN §2.2, §5.6).

	F: forward	B: suffix trie (default)	M: merged, primitive-seeded
Born weighting	at closure	at the leaves (one seed per skeleton class)	at the cut, one seed per colour ordering
Shared	prefixes across families	all prefixes with equal family and remainder	prefixes across families and compositions
Colour certificate	none needed	none needed	certified classes: adjoint rings, traces, quark rings
Role	oracle	default; generic fallback	pure-gluon LC from $n = 6$ (G1); planned default from $n \geq 8$

Colour. Colour lives only in exact seed tables $K[c, b]$, Laurent polynomials in N_c between seed classes c (the colour vector obtained by closing the cut of a skeleton's colour flow) and Born basis elements b; per point the seeds are $U_0 = \sum_b K[c, b]\overline{A_0[b]}$, with a symmetric-group FFT where certified. LC and NLC differ from full colour only in which classes are generated and how cells are truncated: at LC each colour ordering is seeded by its own Born amplitude and skeletons below the leading N_c degree are never generated. Certificates (adjoint rings, fundamental traces, quark rings) are proved at loop level, never inherited from the tree level.

Reduction and masters. Families close directly into NinjaRS `TensorNumerator` lanes: one reduction per (family, sector, column), families with equal (propagators, rank) batched as lanes of one call, scalar masters from oneloopmaster cached per event. A second reduction in a relabelled frame reuses the numerators and the cached masters as the cheap stability check.

Counterterms. The model's UV and R_2 vertices, two-point ones included, are ordinary vertices of the tree recursion, and each contribution contains exactly one of them; external field factors are separate sources. Admission is the requested coupling orders, the exactly-once constraint and a public `particle_veto`: a coupling term whose loop particles include a vetoed particle is dropped, and a vetoed particle is removed from every tree vertex, loop line and propagator of the generation, so loops and counterterms stay consistent. Nothing is matched to the loop (user, 23 October; addendum J). On `loop` the counterterms come from the WP2.2 recursion (one counterterm order in the tree recursion, the default route, whose $gg \to ggggg$ program leaves no rule to site closure) with this admission (`ct-simplify`, b4d76cc3): no witness table and no loop matching. Since A4 (`1b13005a`) the native DAG builder emits this program (byte-identical to the Python builder, which stays as oracle) and builds the loop blocks. The UFO's electroweak counterterms (`392d3f4a`) enter the same way, lowered through the model's own helper bodies with MadLoop's branch side for exactly real arguments, and in 't Hooft–Feynman gauge the Goldstone–vector mixing counterterms as vertex insertions (`8b19dae8`). A counterterm vertex that the model cannot lower and that no tree of the request reaches (the tree engine's own reachability, with the program's order limits and `particle_veto`) is inert and recorded in the artifact; a reachable one is

refused. `particle_veto = [t]` replaces the internal top-free switch of Table 16; site closure and the witness machinery remain only as test oracles.

Precision, stability and assembly. Every kernel is generic over binary64, DoubleFloat, arbitrary-precision floats and exact rationals. A failed lane fails its point and never branches a kernel. The assembly takes $2\,\mathrm{Re}$ of $i/(16\pi^2)$ times the reduced family sum, plus local UV, R_2 and field terms, per sector and column, with optional rotation, reduction-direction and rescaling estimates and the finite-only mode. The public API (TOML, CLI or Python; v2 is the only one-loop engine since MQ-LEGACY-REMOVE) stores plan v4 as a relocatable PACBIN member (zlib JSON of exact kernel term lists, written by the Rust plan model since `79f2a5bd`, no compiled code, digest checked at load) and the native process (`point_inputs = "native-v2"`: the tree program with the Born, counterterm and field components, their saved exact evaluators and the binary64 SymJIT applications of every tree kernel and of the parameter program, stored as IR and recompiled at load) with the companion of the certified-scaleless families and the observables' tables, so no call of a stored process regenerates it (generation-scale-2), on another machine, user and path too; a process the native export cannot store is refused at generation. The Rust/C/C++/Fortran SDK v4 loads the same artifact.

Table 19: **Asymptotic classes per event** for n gluons, full colour (DESIGN §1.4): $F_n \approx (n-1)!\left(\frac{1}{\ln 2}\right)^n/2$ families, 2^n helicities. The required set ($n \le 7$) is far below the crossover of schedule B with the unitarity class ($n \approx 30$–50); schedule M is planned as the default from $n \ge 8$.

Stage or code	Class
Tree blocks	$\Theta(n \cdot n!)$ currents; $\Theta(n! \cdot 2^n)$ current-helicity rows
Dressing, schedule B	$\Theta((n-1)! \cdot 4^n)$ seed injection; trie dressing dominates for $n \le 8$
Dressing, schedule M	$\Theta(n! \cdot 2^n \cdot n^4)$
NinjaRS	$\Theta(F_n)$ reductions, $\Theta(F_n \cdot n^8)$ work, $\Theta(4^n)$ distinct masters after caching
MadLoop (per diagram, per helicity)	$\Theta(D_n \cdot 2^n \cdot n^4)$, D_n super-exponential ($gg \to gggg$: 26 975 loop diagrams)
Unitarity with exact helicity sums (NJet)	$\Theta((n-1)! \cdot 2^n \cdot n^{6\ldots9})$

Table 20: **Ownership of the mathematics.** pyAmpliCol reuses its dependencies for algebra, tensors, graphs and integrals and owns the recursions and the assembly.

Component	Responsibility
ufo_model_loader	Lossless UFO loop-model import: Laurent-valued parameters and couplings, UV/R_2 categories, loop groups
Symbolica	Expressions, coefficient extraction in q, evaluators; exact rationals and arbitrary precision
Spenso, Idenso (Community)	Lorentz and colour tensors behind the kernel catalogue and the colour metrics
FeynKit (Community)	Test oracles: graph and family counts, overall factors, drawings (Figure 3)
NinjaRS, oneloopmaster	Integrand reduction with Laurent expansion; scalar masters in all carriers
pyAmpliCol tree engine	Blocks, Born, coupling orders and colour flow (Python lane; Rust recurrence)
pyAmpliCol loop v2	Families, skeletons, schedules, seed tables, counterterm program, plan v4, native executor, API

8 Sources and reproducibility

- **Build:** `typst compile benchmarks.typ benchmarks.pdf` in `docs/performance_reports/loop/` (Typst 0.15).
- **Status, in-flight work, decisions:** `data/milestone.json`, from the integration merges of e18a024b since de18f169 (onshell-threshold `6e123fdb`, fail-closed-2 `1acb393f`, glibc-floor `5301e24a`, MQ-WARD `d3081568`, MQ-FFT `da6980b0`, MQ-REF `141fe125`, MQ-MODEL `321efdb4`, MQ-STAB `a2aa0d84`, MQ-VALIDATE `01e80d7c`, MQ-NC `d8d9349f`, MQ-5F `6b78dc3b`, MQ-EWJOIN `d525a09c`, MQ-SELECT `e4964fc3`, MQ-LEGACY-REMOVE `539dffdd`, g1-accuracy `e18a024b`; commit subjects "Integrate work/v2-..."); `workspace/reports/engine-v2/slice-status.md` (integration blocks, DECISIONS 2026-10-23 to 29, the milestone pushes m1025a of 2026-10-25 ~13:45 UTC, m1025c of 2026-10-26 ~06:35 UTC (after the m1025b attempt at `db13c260`, not pushed) and m1026a of 2026-10-27 ~21:00 UTC (`8335e9e8`), and milestone m1028a (launched 2026-10-28 ~09:00 UTC; pin `28a16255`, native `loop-native-v2-slice-5cb5ea83`; the user's rules of 2026-10-28: the speed gate at $B = 128$ only (~08:20 item 2, `gate_decisions.gate_batch`), the counterterm-lowering share below 20 % (~08:20 item 1) and the glibc floor (~07:40; 2.38 since glibc-floor); the known limitations from the coordinator's milestone brief and `known_limitations` in `data/g1.json`; the identical-flavour LC limitation of m1026a, DECISIONS 2026-10-26 ~21:10 item 1, is removed by identical-flavour-lc) and milestone M-QCD (DECISIONS 2026-10-29 ~17:20: pin `e18a024b`, native `loop-native-v2-slice-d0fef168` the user's milestone plan of 2026-10-28 ~12:50 and its rulings ~15:25, ~15:35/40 and ~16:10, the user's rulings of 2026-10-29 ~03:45 (E1) and ~10:20 (precision), the coordinator's ~11:25 (F7, F11) and R3 for QCD ~18:25; `workspace/reports/engine-v2/m-qcd/PLAN.md` §1, §5); the user's gate decision of 2026-10-24, 16:05 UTC, is recorded in `data/milestone.json` `gate_decisions` and no longer applies after the re-measure; the user's rules of 2026-10-24, 22:35 UTC — uncoloured generation, green-to-red run times, summary pages with only Table 1, Figure 1 and legends, and the gate rule "below 5 ms: not gated (user 2026-10-24)", recorded as `gate_decisions.threshold_rule` and applied by `scripts/build_summary.py` and this report); `HANDOFF_24_10_2026.md` §6; MASTER_PLAN 2026-10-23/24 entries; REQUIREMENTS_ADDENDUM A, D, F, G, H, I, J; DESIGN §5.5, §3.3, §5.7 (G1), §6.1, §6.2, §7.2, §7.5, §7.9, §12.1, §12.4, WP3.1, WP3.2; the 24

October pushes (59badc7a and b4d76cc3, gates in slice-status.md), the G-int push (79df367a, 25 October), the m1025a push (bae3b9ea, 25 October), the m1025c push (36f43250, 26 October) and the m1026a push (8335e9e8, 27 October). WP3.2 increments and dependencies: `workspace/reports/engine-v2/wp32/PLAN.md`; the EW counterterm comparison with MadLoop and measurement M1: `wp32/M1-ew-ct.md` and DESIGN §6.1; the genericity audits: `workspace/reports/engine-v2/genericity-audit/REPORT.md` and `genericity-audit-2/REPORT.md` (at `11542fd5`, with the first audit's follow-up); precision-ap's MadLoop comparison: `workspace/reports/engine-v2/precision-ap/`; reference-relocate's inventory and evidence: `workspace/reports/engine-v2/reference-relocate/`; model-determinism: DESIGN WP2.5; native-lowering: DESIGN §12.1 and WP3.1 (Rust migration D); onshell-consistency: DESIGN §11.1 and `docs/user/configuration.md`; native-seeds and native-cells: DESIGN §7.2, §7.9 and `workspace/reports/engine-v2/wp31/RUST-MIGRATION.md`; fermion-lines: DESIGN §3.3 and the fixtures `tests/fixtures/loop/required/uux_{ddxssx,uuxddx,uuxuux,uuxepem}.json`; generation-scale-2: DESIGN WP3.1 (its status) and §12.1; EWI-4b and EWI-6b: DESIGN §12.4 and `wp32/PLAN.md`; the packages of m1028a: their integration blocks in slice-status.md (ew-ct EWI-3R `59d47cb3`, ew-epsilon EWI-7b `6da37352`, fail-closed `fb54246c`, sdk-parity `f6619f34`, ew-leptons EWI-5 `b82d8280` with its R2_G0tt_b evidence (DECISIONS 2026-10-28 ~05:20 item 2 and ~08:10 item 1), identical-flavour-lc `5cb5ea83`, gate-b128 `28a16255`), DESIGN §5.7 ("Gate batch"), §12.4 and the listed generation refusals, `docs/user/configuration.md` and `docs/user/ufo-model-coverage.md`; the packages of M-QCD: their integration blocks in slice-status.md (Section 1), `docs/user/loop-sm-and-madloop.md` (the `loop_sm` repair rule) and the listed refusals of `src/pyamplicol/loops/v2/refusals.py` and `processes/refusals.py`.

- **Section 1 (M-QCD), Section 2, Table 3, Table 21:** `data/mqcd.json` (`scripts/build_mqcd.py` from the campaign's records), `data/mqcd-findings.json` (MQ-VALIDATE's triage, re-checked at the pin), `data/mqcd-public.json` (the public-route representatives), `data/mqcd-fft-cost.json` and `data/mqcd-fft-worst.json` (FFT cost and worst points), `data/mqcd-selftest.json` (wheel self-test), combined by `scripts/build_mqcd_matrix.py` into `data/mqcd-matrix.json`; the campaign's records, logs and scripts in `workspace/reports/engine-v2/m-qcd/records/m1029-mqcd/` (a copy of `workspace/tmp/campaign-m1029-mqcd`: `c1/` results, JUnit and logs, `x/` the FFT and stock-`loop_sm` test logs, `public/`, `fft/`, `wheel/`, `STATE.md`).
- **Table 1, Figure 1:** `data/summary.json`, written by `scripts/build_summary.py` from `data/g1.json`, `data/measured.json`, `data/milestone.json` and `tools/loop_benchmark/processes.json` by its main-configuration rule (the most recent measured configuration of each cell; a newer check that only repeats part of an older comparison does not replace it; revision ancestry from `data/milestone.json`); every entry names its configuration and source (README).
- **Table 13:** HANDOFF §0.1 (addendum F); the provider generation records of Table 1 (`data/summary.json`): `data/g1.json` for the G1 rows and the reference runs' records (`workspace/reference/runs/required/<process>/`, collected into `data/measured.json`) for the $2 \to 5$ rows; MadLoop's production-build times of the $2 \to 5$ rows: the same runs' `madloop-production-result.json`. **Table 14, Section 4, Section 5:** the native validations on `work/v2-generation-scale`, records `workspace/tmp/generation-scale/gint2/after/`, `gint/val1/` and `i7/val/` (the earlier seven-gluon points: `gint/val1/`), collected into `data/measured.json`; the earlier $gg \to t\bar{t}e^-e^+g$ record: DESIGN WP3.1 fourth increment. **Conventions:** MadLoop standalone, Recola, OpenLoops 2.
- **Section 3 (Table 4 to Table 12):** `data/g1.json` and the per-point values `data/g1-values.json`, written by `python -m tools.loop_benchmark gate` (README and docstrings of `tools/loop_benchmark`); streams `tools/loop_benchmark/streams/`; run records archived in `workspace/reports/engine-v2/measure-g1/records/` (first campaign `runs/`, fix round `runs2/`, the re-measure at G-int (`4e9b5a9e`) `gint/` with its paired binary64-input runs `confirm/<process>/ab.json`, milestone m1025a's at `1d27426b` `m1025a/` (with the frozen-heap launcher `gc_launcher.py` and its protocol experiment `gc/`), milestone m1026a's at `11542fd5` `m1026a/` (a copy of `workspace/tmp/g1-m1026a/`, with the cold/warm generation A/B `ab/` and in `killed/` the first full-colour seven-gluon run, no gate input), milestone m1028a's at `28a16255` `m1028a/` (a copy of the campaign `workspace/tmp/g1-m1028a/`: `runs/` with every row's timing, generation, accuracy check and stability rows, round 2 of the $2 \to 2$ to $2 \to 4$ rows (`m1028a-r2-*`, deciding) and the seven-gluon stores' records, `logs/` with `/usr/bin/time -v` and host-load samples (`hostload.txt`, `hostload-r2.txt`), the public-path confirmation `confirm/`, the same-core old/new A/B and bisect runs `ab/` with `logs/ab-summary.txt`, the gate inputs `out/inputs-final.txt` and labels `out/meta-final.json`, the campaign's scripts `scripts/`) and milestone M-QCD's at `e18a024b` `m1029-mqcd/` (a copy of `workspace/tmp/g1-m1029-mqcd`: `runs/`, `logs/`, `bin/`, the gate inputs `inputs-final.txt`, labels `meta-final.json`, `compare.json`, `gate.log`, `STATE.md`; the seven-gluon stores are not copied); provider records reused from the first campaign and the fix round, not re-run); the earlier $2 \to 5$ records and the provider runs of those rows in `workspace/reports/engine-v2/measure-g1-7leg/records/` (`runs-f2/`, `runs-f4/`), MadLoop exports of the G1-only processes in `workspace/reference/runs/g1/`; recalibration `tools/loop_v2_costmodel.py --g1 data/g1.json`. Builds: `loop-native-v2-slice-d0fef168` (`loop`, `e18a024b`); history: `loop-native-measure-g1-genscale-39824f9b` (`work/v2-generation-scale` `889af02c` plus the timers of scratch commit `39824f9b`, tag `g1-native-inputs-39824f9b`: earlier runs, one lane, schedule M, stability) and `loop-native-measure-g1-033edf15` (Python inputs); the earlier runs of m1028a: `loop-native-v2-slice-5cb5ea83`, of m1026a: `loop-native-v2-slice-11542fd5`, of m1025a: `loop-native-v2-slice-7db32901`, of G-int: `loop-native-v2-slice-4e9b5a9e`, release with one codegen unit and thin LTO; the harness is the committed `tools/loop_benchmark` of `e18a024b` for every row (the heap freeze in the harness; m1025a's launcher and `measure-g1-7leg`'s run worktree `2265a9f7` are history). Schedule M at seven legs: DESIGN §5.6.
- **Table 15** and NJet prose: `workspace/reports/engine-v2/njet-benchmark.md` §2, §5, §6 (raw `workspace/reference/runs/njet/timing/`; the seven-gluon single evaluations are read from those records into `data/measured.json`); NJet, arXiv:1209.0100 Table 5.
- **Table 16:** `data/measured.json` (`scripts/collect_milestone.py <workspace>`); top-free values at lower multiplicity: DESIGN WP3.1 and `test_loop_v2_topfree.py`; the seven-gluon generation history: DESIGN WP3.1 eighth and ninth increments. Memory in GiB = 2^{30} bytes.
- **Table 17, Section 7:** `docs/development/loop_v2/DESIGN.md` §1–§3, §5.5, §5.6, §6, §10, §12, REQUIREMENTS_ADDENDUM J (counterterm rule) and the integration records above.
- **History:** the legacy per-diagram engine's snapshot of 2026-10-18 (`data/measurements.json`, `provider-cold.json`, `legacy-labels.json`, `colour-measurements.json`) stays for the DESIGN citations and is no longer shown; its table is in the earlier report revisions (README).

Appendix: the M-QCD campaign rows

Table 21: **Rows of the M-QCD validation campaign** (data/mqcd.json, written by scripts/build_mqcd.py from the campaign's records; e18a024b (M-QCD pin, native loop-native-v2-slice-d0fef168). Per row: class, process and loop_sm restriction, tiers (d default, n nightly) with their points, the reference providers of the row and its gating variants (ML MadLoop, Rc Recola, OL OpenLoops, NJet; MadLoop in quadruple precision from MQ-REF), the largest deviations over the row and its gating variants (DD from the reference, rule 1; binary64 from DD where asserted, rule 2; poles from Catani in DD, rule 4; — none formed), the extras and gating variants (status where not pass), and the row's status; the failures and their findings are in data/mqcd.json and data/mqcd-findings.json.

Row	Class	Process	Restriction	Points	Reference	DD	b64/DD	Catani	Extras · variants	Status
A01	Q-massless	$gg \to gg$	no_b_mass	d 2 · n 1	ML, Rc, OL	1.1e−15	2.9e−14	1.2e−31	nc, ap · sectors	pass
A02	Q-massless	$u\bar{u} \to gg$	no_b_mass	n 3	ML	1.2e−14	4.2e−13	1.4e−31	madloop	pass
A03	Q-massless	$gg \to u\bar{u}$	no_b_mass	n 4	ML	6.6e−15	4.2e−14	1.6e−31		pass
A04	Q-massless	$ug \to ug$	no_b_mass	d 2 · n 2	ML	1.0e−15	7.3e−15	2.2e−30		pass
A05	Q-massless	$u\bar{u} \to d\bar{d}$	no_b_mass	n 4	ML	1.1e−15	6.8e−15	1.9e−31	uv	pass
A06	Q-massless	$u\bar{u} \to u\bar{u}$	no_b_mass	n 3	ML, Rc, OL	2.8e−14	5.8e−13	1.1e−28	sectors	pass
A07	Q-massless	$uu \to uu$	no_b_mass	n 4	ML	2.6e−15	2.9e−14	1.6e−30		pass
A08	Q-massless	$gg \to ggg$	no_b_mass	n 4	ML	3.4e−14	4.5e−13	6.1e−30	nc, ward, hel	pass
A09	Q-massless	$u\bar{u} \to ggg$	no_b_mass	d 2 · n 2	ML, Rc, OL	7.8e−13	4.5e−11	1.2e−22	nc, sp · soft, collinear fail	fail
A12	Q-massless	$u\bar{u} \to d\bar{d}g$	no_b_mass	d 2 · n 1	ML, Rc, OL	5.0e−14	4.5e−13	7.7e−30	nc · sectors	pass
A13	Q-massless	$u\bar{u} \to u\bar{u}g$	no_b_mass	n 3	ML, Rc, OL	6.3e−14	7.9e−13	5.3e−30	nc, ap · sectors	pass
A14	Q-massless	$gg \to gggg$	no_b_mass	n 3	ML, Rc, OL	3.3e−14	5.0e−11	1.4e−23	ward	pass
A15	Q-massless	$u\bar{u} \to gggg$	no_b_mass	n 4	ML	3.4e−14	4.6e−12	1.7e−29	nc	pass
A18	Q-massless	$gg \to u\bar{u}u\bar{u}$	no_b_mass	n 4	ML, Rc, OL	1.2e−11	1.1e−10	1.4e−27	nc	pass
A19	Q-massless	$u\bar{u} \to d\bar{d}s\bar{s}$	no_b_mass	n 3	ML, Rc, OL	8.4e−16	2.0e−13	8.1e−30	sectors	pass
A20	Q-massless	$u\bar{u} \to u\bar{u}d\bar{d}$	no_b_mass	n 3	ML, Rc, OL	1.9e−14	2.4e−12	2.5e−30	sectors	pass
A21	Q-massless	$u\bar{u} \to u\bar{u}u\bar{u}$	no_b_mass	n 3	ML, Rc, OL	3.7e−14	4.1e−12	2.3e−29	nc · sectors	pass
A22	Q-massless	$u\bar{u} \to d\bar{d}s\bar{s}g$	no_b_mass	n 3	ML, Rc	7.0e−13	4.8e−13	1.3e−28		pass
A23	Q-massless	$u\bar{u} \to ggggg$	no_b_mass	n 3	ML, Rc	3.2e−12	1.4e−12	2.7e−29		pass
A24	Q-massless	$gg \to ggggg$	no_b_mass	n 2	NJet	8.0e−17	3.3e−12	—	uv	partial
A25	Q-massless	$ud \to ud$	no_b_mass	n 4	ML	2.1e−15	5.8e−14	9.2e−30		pass
A26	Q-massless	$\bar{u}\bar{u} \to \bar{u}\bar{u}$	no_b_mass	n 4	ML	2.6e−15	8.4e−14	1.6e−30		pass
B01	Q-top	$gg \to t\bar{t}$	no_b_mass	d 2 · n 1	ML, Rc, OL	1.7e−15	1.2e−10	3.0e−26	ap, uv, hel, sp · threshold, mt170, sectors	pass
B02	Q-top	$u\bar{u} \to t\bar{t}$	no_b_mass	n 3	ML	3.9e−14	8.7e−14	4.0e−30	madloop	pass
B03	Q-top	$gg \to t\bar{t}g$	no_b_mass	n 4	ML	3.4e−13	3.4e−11	7.5e−27	nc, ward	pass
B05	Q-top	$gg \to t\bar{t}gg$	no_b_mass	n 3	ML, Rc, OL	1.3e−13	3.2e−11	1.4e−27	sectors	pass
B06	Q-top	$u\bar{u} \to t\bar{t}d\bar{d}$	no_b_mass	n 4	ML	1.5e−14	1.5e−12	1.8e−28		pass
B07	Q-top	$gg \to t\bar{t}t\bar{t}$	no_b_mass	n 4	ML	2.5e−14	1.4e−13	1.4e−29		pass
B08	Q-heavy	$gg \to t\bar{t}$	default	n 4	ML	1.9e−15	3.1e−14	1.5e−30		pass
C01	Q-heavy	$gg \to b\bar{b}$	default	d 2 · n 2	ML, Rc	9.3e−16	6.2e−15	8.1e−29	nc, ap, ward, sp · params	pass
C02	Q-heavy	$gg \to gg$	default	n 4	ML	5.0e−14	9.0e−12	1.7e−29	uv	pass
C03	Q-heavy	$b\bar{b} \to h$	default	n 4	ML	3.9e−16	4.3e−16	9.9e−31		pass
C04	Q-heavy	$b\bar{b} \to hg$	default	n 4	ML	5.9e−16	1.9e−15	3.2e−30		pass
C05	Q-heavy	$gg \to b\bar{b}h$	default	n 4	ML	5.0e−15	2.3e−10	1.6e−29		pass
C06	Q-heavy	$gg \to t\bar{t}b\bar{b}$	default	n 4	ML	1.9e−15	1.5e−13	1.9e−29		pass
C07	Q-singletop	$ug \to dt\bar{b}$	default	n 4	ML	8.0e−15	1.6e−14	—		pass
C08	Q-heavy	$gg \to c\bar{c}$	c_mass	n 4	ML, Rc	1.7e−15	4.5e−15	4.9e−29		pass
C09	Q-heavy	$gs \to w^- c$	c_mass	n 4	ML, Rc	4.6e−16	1.5e−14	1.1e−28		pass
D01	Q-EWBorn	$u\bar{u} \to e^- e^+$	no_b_mass	d 2 · n 2	ML	5.3e−16	4.9e−15	1.0e−31	ap, ff	pass
D02	Q-EWBorn	$u\bar{u} \to e^- e^+$	no_b_mass	n 4	ML	7.1e−16	4.3e−15	2.1e−31		pass
D05	Q-EWBorn	$u\bar{u} \to e^- e^+ g$	no_b_mass	d 2 · n 2	ML	2.3e−14	2.5e−14	1.6e−31	nc, ward, uv, hel	pass
D06	Q-EWBorn	$u\bar{u} \to e^- e^+ gg$	no_b_mass	n 3	ML, Rc	8.6e−14	1.1e−12	2.7e−31	madloop	pass
D07	Q-EWBorn	$u\bar{u} \to e^- e^+ ggg$	no_b_mass	n 3	ML, Rc, OL	1.3e−14	5.2e−13	2.4e−30	sectors	pass
D08	Q-EWBorn	$u\bar{u} \to e^- e^+ d\bar{d}g$	no_b_mass	n 3	ML, Rc, OL	1.1e−14	8.9e−13	1.4e−30	sectors	pass
D09	Q-EWBorn	$u\bar{u} \to e^- e^+ m^- m^+$	no_b_mass	n 3	ML	8.9e−16	2.7e−14	5.3e−31	sectors	pass
D10	Q-EWBorn	$u\bar{u} \to u\bar{u}e^+ e^-$	no_b_mass	n 3	ML, Rc	3.1e−15	1.7e−12	3.0e−30	sectors	pass
D11	Q-EWBorn	$u\bar{u} \to zg$	no_b_mass	n 4	ML	7.1e−15	3.4e−14	2.5e−30		pass
D12	Q-EWBorn	$u\bar{d} \to e^+ \text{ ve } g$	no_b_mass	n 4	ML	1.3e−15	1.2e−14	1.4e−31		pass
D13	Q-EWBorn	$ug \to au$	no_b_mass	n 4	ML	1.2e−15	1.2e−14	4.9e−31		pass
D15	Q-EWBorn	$u\bar{u} \to zz$	no_b_mass	n 4	ML	3.2e−15	2.8e−13	1.5e−30		pass
D16	Q-EWBorn	$u\bar{u} \to w^+ w^-$	no_b_mass	n 4	ML	1.2e−13	1.4e−12	3.2e−28		pass
D17	Q-EWBorn	$b\bar{b} \to w^+ w^-$	default	n 4	ML	4.6e−14	6.6e−13	5.7e−29	5F	pass
D18	Q-EWBorn	$u\bar{u} \to zh$	no_b_mass	n 4	ML	8.5e−16	2.0e−12	1.1e−29		pass
D19	Q-EWBorn	$ud \to udh$	no_b_mass	n 4	ML	5.9e−16	3.3e−15	—		pass
D20	Q-EWBorn	$e^+ e^- \to u\bar{u}g$	no_b_mass	n 4	ML	7.3e−14	2.0e−12	7.0e−30		pass
D21	Q-EWBorn	$e^+ e^- \to t\bar{t}$	no_b_mass	n 3	ML	1.3e−14	1.8e−14	1.2e−31		pass
D23	Q-EWBorn	$ag \to u\bar{u}$	no_b_mass	n 4	ML	4.3e−16	3.1e−13	3.2e−31		pass
D24	Q-EWBorn	$e^- u \to e^- u$	no_b_mass	n 4	ML	4.8e−16	4.1e−15	8.0e−32		pass
D25	Q-EWBorn	$u\bar{d} \to w^+ a$	no_b_mass	n 4	ML	4.8e−15	4.4e−13	1.3e−28	sp · zero	pass
D26	Q-heavy	$u\bar{u} \to e^- e^+ g$	default	n 4	ML	2.3e−14	8.5e−14	1.5e−31		pass
D27	Q-EWBorn	$gu \to e^+ \text{ ve } d$	no_b_mass	n 4	ML	1.3e−15	4.3e−14	6.5e−31		pass
D28	Q-EWBorn	$u\bar{u} \to w^+ w^- g$	default	n 4	ML	1.8e−12	1.4e−11	2.6e−28	ward	pass

Row	Class	Process	Restriction	Points	Reference	DD	b64/DD	Catani	Extras · variants	Status
D29	Q-EWBorn	$b\bar{b} \to w^+w^-g$	default	n 4	ML	1.7e−13	8.8e−12	4.5e−28	ward · 5F	pass
E01	Q-ttX	$gg \to t\bar{t}h$	no_b_mass	n 4	ML	3.6e−13	7.4e−12	6.7e−30	nc, ward, uv	pass
E02	Q-ttX	$gg \to t\bar{t}z$	no_b_mass	n 4	ML	7.2e−14	3.2e−12	5.8e−30		pass
E05	Q-ttX	$e^+e^- \to t\bar{t}h$	no_b_mass	n 3	ML	5.8e−15	6.6e−15	2.4e−31	ap	pass
E06	Q-ttX	$gg \to t\bar{t}e^-e^+g$	no_b_mass	n 3	ML, Rc, OL	6.7e−14	4.4e−12	5.7e−27	sectors	pass
F01	Q-singletop	$ub \to dt$	default	n 4	ML	1.2e−15	8.7e−16	1.3e−31	nc, uv, ap · 5F	pass
F02	Q-singletop	$bu \to dt$	default	n 4	ML	1.2e−15	1.8e−15	1.1e−31	5F	pass
F03	Q-singletop	$u\bar{d} \to t\bar{b}$	no_b_mass	d 2 · n 2	ML	1.7e−15	4.1e−16	8.6e−32		pass
F04	Q-singletop	$gb \to tw^-$	default	n 4	ML	6.2e−16	3.2e−15	3.4e−30	ward · 5F	pass
G01	Q-mixed	$u\bar{u} \to d\bar{d}$	none	d 2 · n 1	ML	4.6e−15	2.4e−15	5.0e−32	uv, ap	pass
G02	Q-mixed	$u\bar{u} \to t\bar{t}$	no_b_mass	n 4	ML	2.0e−15	3.1e−14	3.7e−30		pass
H01	Q-CKM	$u\bar{s} \to w^+g$	zeromass_ckm	n 4	ML	1.4e−15	7.2e−14	4.6e−29	uv, ap	pass
H02	Q-CKM	$gs \to tw^-$	zeromass_ckm	n 4	ML	1.2e−15	7.6e−15	2.8e−30		pass
H03	Q-CKM	$gd \to tw^-$	ckm	n 4	ML	1.2e−15	5.6e−15	4.5e−30		pass
M01	Q-multi	$pp \to t\bar{t}$	no_b_mass		ML	2.0e−15	4.1e−14	4.0e−30	P0, public-P0, P2, public-P2	pass
R01	Q-EWBorn	$u\bar{u} \to m^-m^+$	lepton_masses	n 4	ML	3.1e−16	1.2e−15	3.9e−32		pass